GHXTox: Peptide toxicity prediction by fusing a task-adaptive protein language model with pLDDT-aware multi-scale 3D structural modeling

LaLaLa, LaLaLa, LaLaLa, LaLaLa

Abstract

Peptide toxicity is a major safety concern in peptide-based drug development, motivating accurate computational screening methods. Existing methods have begun to explore multi-dimensional (multimodal) feature fusion and toxicity prediction based on the three-dimensional (3D) conformations of peptides, yet most of them still underutilize 3D structural information, and systematic multi-dimensional fusion remains rare. Here, we propose GHXTox, a hierarchical dual-level fusion framework in which a 3D structural expert first integrates sequence semantics with spatial structural features through predicted local-distance difference test (pLDDT)-aware cross-attention, and is then fused at the probability level with a task-adaptive one-dimensional (1D) sequence expert using weights frozen on training out-of-fold (OOF) predictions. The 1D branch analyzes sequences with a protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) and a bidirectional Mamba state-space model. The 3D expert integrates an ESMFold-derived structural anchor, pLDDT-aware residue geometry, and full-atom-derived chemical sites. Multi-conformer information is retained only in training-side representation learning and controlled ablations; benchmark and public inference always use the single-structure base logits. On the frozen historical benchmark test sets, GHXTox achieves MCC values of 0.8705 and 0.6673 and AUPRC values of 0.9674 and 0.8046 on Test1 and Test2, respectively. Further analyses characterize residue-level sequence sensitivity, confidence–gate associations, chemical-site perturbations, expert disagreement, and an OOF-frozen selective-prediction rule. At realized coverages of 88.54% and 86.94%, the selective rule reduced the accepted-sample error to 2.71% and 3.16% on Test1 and Test2 without benchmark-specific cutoff selection. These findings are interpreted as model-level evidence rather than causal biological mechanisms. Frozen-model analyses indicate that the conformer contribution is small and non-uniform: under the structural expert, substantial geometric variation does not necessarily translate into stable outcome-aligned probability variation.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware multi-scale 3D modeling; interpretability

1 INTRODUCTION

Peptides are increasingly used as therapeutic candidates because of their high target specificity and functional diversity [1]. However, peptide toxicity remains a major safety concern during drug development, and the accurate and efficient identification of toxic peptides is a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable.

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional,

1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks, and a bidirectional long short-term memory network with residual connections, achieving strong performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, such methods are constrained by the limited accuracy of current structure prediction models on short peptides and typically rely on a single predicted conformation; structural confidence, i.e., the predicted local distance difference test (pLDDT) score, is either discarded through hard filtering or ignored, and multi-conformer ensembles are not considered. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines a task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable hierarchical dual-level fusion framework for peptide toxicity prediction. A task-adaptive 1D sequence expert combines a ProtT5 protein language model, adapted by low-rank adaptation (LoRA), with a bidirectional Mamba state-space encoder and three-seed ensembling [8–10]. A sequence-enhanced 3D structural expert integrates sequence semantics with spatial structure through pLDDT-aware cross-attention, ESMFold-derived geometry, full-atom chemical sites, and confidence gating across multiple structural scales; PepFlow and AlphaFold2/ColabFold conformer ensembles informed only training-side analyses and controlled ablations, and the final predictor uses the single-structure ESMFold anchor [11–14]. The two expert probabilities are fused at the probability level with weights fixed on training out-of-fold (OOF) predictions; the 1D probabilities are first calibrated on the same OOF table, and no benchmark labels are used during this selection. Gated sequence–structure cross-attention inside the structural expert further informs the interaction design [15]. Ablation experiments show that the contribution of structural evidence is dataset- and metric-dependent: fusion improves several fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2. Post hoc analyses characterize residue-level sensitivity, confidence gating, chemical-site perturbations, and expert disagreement without assigning causal biological meaning to these signals. The framework is evaluated on the frozen historical benchmarks described in Section 2.1.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable hierarchical dual-level fusion framework that integrates a task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling; (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on the frozen historical benchmark test sets show that GHXTox improves several fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2, while ablation, robustness, and interpretability analyses characterize

the conditions under which sequence and structural evidence are complementary.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [16] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [17], ToxinPred2 [18], and ATSE [19], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [20] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [21] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as the benchmark Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional benchmark Test2 set, containing 46 toxic and 536 non-toxic peptides, is used for frozen historical evaluation. Both Test1 and Test2 had been observed during historical project development and are therefore treated as frozen historical benchmarks rather than newly blinded external validation sets; their labels were not used in the formal OOF calibration, fusion-weight selection, or threshold-freezing procedure. ToxinPred 3.0 [22] was consulted as an external method and data-source reference during project development, but its records were not added to the present training set or to the formal Test1/Test2 result tables. Table 1 summarizes the statistics of the datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Benchmark test
Test2	46	536	582	Benchmark test

2.2 Overview of the Model

GHXTox is a hierarchical dual-level fusion framework for peptide toxicity prediction: a task-adaptive one-dimensional (1D) sequence expert and a sequence-enhanced three-dimensional (3D) structural expert generate independent probabilities, interact internally at the feature level, and are combined at the probability level.

Figure 1: Overall architecture of GHXTox. The framework combines a task-adaptive 1D sequence expert with a pLDDT-aware structural expert through probability-level fusion. The conformer residual is restricted to training-side analysis and is disabled during benchmark and public inference.

The 1D expert processes the peptide sequence with a LoRA-adapted ProtT5 encoder [8, 9] followed by a bidirectional Mamba state-space model [10], averages three seed models, and applies OOF-fitted Platt calibration to produce p_{1D} (Section 2.3). The 3D expert integrates sequence semantics, residue geometry, full-atom chemical sites, and pLDDT confidence at five

scales, combines sequence and spatial states through pLDDT-aware cross-attention, and emits p_{SE3D} (Section 2.4).

The two expert probabilities are fused linearly in probability space with weights fixed on training out-of-fold (OOF) predictions; the 0.72/0.28 weights are an empirical OOF-selection outcome, not a manually assigned prior:

$$p_{\mathrm{GHXTox}} = 0.72\, p_{\mathrm{1D}} + 0.28\, p_{\mathrm{SE3D}}. \tag{1}$$

The fused probability is converted into a binary prediction with the same OOF-fixed threshold:

$$\hat{y} = \begin{cases} 1, & p_{\mathrm{GHXTox}} \geq 0.4702225653, \\ 0, & p_{\mathrm{GHXTox}} < 0.4702225653, \end{cases} \tag{2}$$

where $\hat{y} = 1$ denotes a toxic peptide and $\hat{y} = 0$ a non-toxic peptide. These parameters are applied unchanged to both benchmarks (Section 2.5.4).

2.3 Task-Adaptive 1D Sequence Modeling

The 1D sequence expert is designed to learn toxicity-related contextual semantics and sequential dependencies directly from peptide amino acid sequences. Instead of reducing a frozen protein-language-model output to a fixed mean vector, GHXTox preserves the residue-level representations generated by ProtT5 and adapts the pretrained encoder to the peptide toxicity prediction task through low-rank adaptation (LoRA). The resulting residue-level representations are then processed by a bidirectional Mamba state-space encoder to jointly model local residue patterns, N-to-C dependencies, and C-to-N dependencies. The encoded residue features are aggregated via masked mean pooling and passed to a linear classification head to produce the single-model toxicity probability. Finally, three models trained on the full training set with different random seeds are combined by equal-weight probability averaging, followed by Platt calibration fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability p_{1D}.

2.3.1 Residue Tokenization and LoRA-Adapted ProtT5

Given a peptide sequence of length L, residues are converted into the space-separated format required by the ProtT5 tokenizer and mapped to token indices; sequences of different lengths are padded within each batch. A binary residue-validity mask distinguishes residue positions from special and padded tokens and is applied in the Mamba scans and subsequent pooling.

The tokenized sequence is processed by the ProtT5-XL-UniRef50 encoder [8] with LoRA, producing residue-level representations $\mathbf{H}^{\mathrm{PLM}} \in \mathbb{R}^{L\times 1024}$. Retaining the complete residue-level representation matrix preserves position-specific contextual semantics for the downstream encoder; each residue vector encodes both its amino acid identity and its context within the peptide sequence.

To adapt the general-purpose encoder to the toxicity task, low-rank adaptation (LoRA) [9] is applied to the linear transformations within the encoder. For an original linear layer $\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b}$, the pretrained weight matrix $\mathbf{W}_0$ is frozen, and the task-specific update is factorized into two trainable low-rank matrices:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \qquad \mathbf{A} \in \mathbb{R}^{r\times d_{\mathrm{in}}}, \quad \mathbf{B} \in \mathbb{R}^{d_{\mathrm{out}}\times r}. \tag{3}$$

We set the LoRA rank to $r = 8$, the scaling coefficient to $\alpha = 16$, and the LoRA dropout rate to 0.1, yielding a scaling factor of $\alpha/r = 2$. Matrix $\mathbf{A}$ is initialized with Kaiming initialization, whereas $\mathbf{B}$ is initialized to zero, so the LoRA branch initially leaves the pretrained transformation unchanged. In contrast to configurations that apply LoRA only to selected attention projections,

all linear layers within the ProtT5 encoder are augmented; the pretrained base transformations and embeddings remain frozen, whereas the low-rank parameters and the downstream sequence-classification modules are trainable.

The 1024-dimensional ProtT5 representations are then projected into the 192-dimensional hidden space used by the Mamba encoder [10]:

$$\mathbf{X}^{(0)} = \mathrm{Dropout}\left(\mathrm{GELU}\left[\mathrm{LN}\left(\mathbf{H}^{\mathrm{PLM}}\mathbf{W}_p + \mathbf{b}_p\right)\right]\right), \qquad \mathbf{X}^{(0)} \in \mathbb{R}^{L\times 192}.$$

where LN denotes layer normalization.

2.3.2 Bidirectional Mamba Sequence Encoder

Although ProtT5 provides contextualized residue representations, these representations primarily encode general protein semantics learned during large-scale pretraining. To further model the sequential dependencies specifically associated with peptide toxicity, two bidirectional Mamba layers are placed after the ProtT5 projection block. Each directional block follows the standard selective state-space formulation of Mamba [10]: the input is projected into content and gating branches, processed by a depthwise convolution, discretized with input-dependent step sizes, and updated through a recurrent state of dimension 16. In compact form, the discretized transition and the state update are

$$\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A}), \qquad \mathbf{S}_t = \bar{\mathbf{A}}_t \odot \mathbf{S}_{t-1} + \bar{\mathbf{B}}_t \odot \widetilde{\mathbf{U}}^{\mathrm{m}}_t, \tag{4}$$

followed by a gated output projection. At padding positions, the convolved feature and the step size are zeroed, preventing padding tokens from injecting information into the recurrent states.

A unidirectional scan accesses only one side of the sequence context, whereas peptide toxicity may depend on residue patterns associated with both termini and with interactions spanning different sequence positions. In the ℓ-th bidirectional layer, forward and backward scans are therefore applied to the normalized sequence using the residue-validity mask $\mathbf{m}$, and the two directional outputs are concatenated and fused by a linear layer with a residual connection:

$$\begin{aligned}
\mathbf{F}^{(\ell)} &= \mathcal{M}_f^{(\ell)}(\mathrm{LN}(\mathbf{X}^{(\ell)}), \mathbf{m}), \\
\mathbf{R}^{(\ell)} &= \mathrm{Reverse}\left[\mathcal{M}_b^{(\ell)}\left(\mathrm{Reverse}\left[\mathrm{LN}(\mathbf{X}^{(\ell)})\right], \mathrm{Reverse}(\mathbf{m})\right)\right], \\
\mathbf{X}^{(\ell+1)} &= \mathbf{X}^{(\ell)} + \mathrm{Dropout}\left(\mathrm{GELU}\left(\mathbf{W}_m\left[\mathbf{F}^{(\ell)} \parallel \mathbf{R}^{(\ell)}\right] + \mathbf{b}_m\right)\right),
\end{aligned} \tag{5}$$

where $\parallel$ denotes feature concatenation and the backward-scan output is restored to the original residue order. Two such layers are stacked in the final model, allowing each residue position to receive conditional information from both its N-terminal and C-terminal contexts.

2.3.3 Masked Pooling and Class-Weighted Classification

After two bidirectional Mamba layers, the model produces residue-level representations $\mathbf{X}^{(2)} \in \mathbb{R}^{L\times 192}$. Because peptide sequences have variable lengths, GHXTox does not rely on a fixed-position classification token. Instead, masked mean pooling is applied over the valid amino acid positions:

$$\mathbf{h}_{1D} = \frac{\sum_{i=1}^{L} m_i \mathbf{x}_i}{\sum_{i=1}^{L} m_i}, \tag{6}$$

where the denominator equals the number of valid residue positions. This operation converts the variable-length residue sequence into a fixed 192-dimensional peptide-level representation while ensuring that EOS and padding positions do not contribute to the pooled feature; unlike global max pooling, it aggregates distributed evidence across the entire peptide.

The pooled representation is normalized and passed through a single-output linear classification head, $z_{1D} = \mathbf{w}_c^\top \mathrm{LN}(\mathbf{h}_{1D}) + b_c$, giving the uncalibrated toxicity probability $p_{1D}^{\mathrm{raw}} = \sigma(z_{1D})$. The classification head therefore consists of layer normalization followed by a single linear layer, rather than a multilayer perceptron.

Because non-toxic peptides substantially outnumber toxic peptides in the training data, a positive-class-weighted binary cross-entropy loss is used in the current 1D expert instead of SMOTE or Borderline-SMOTE. For a training partition containing N_+ toxic and N_- non-toxic peptides, the positive-class weight is $w_+ = N_-/N_+$, and the class-weighted objective is

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N}\sum_{j=1}^{N}\left[w_+ y_j \log\sigma(z_j) + (1-y_j)\log(1-\sigma(z_j))\right]. \tag{7}$$

During five-fold out-of-fold training, w_+ is calculated independently from the class counts of each fold's training partition; for full-data training, it is recalculated using the complete training set. This strategy increases the contribution of the minority toxic class without generating synthetic peptide sequences or feature vectors, while preventing any information from the held-out validation fold from entering the training procedure.

2.3.4 Three-Seed Ensemble and OOF Platt Calibration

The predictions of deep neural networks may vary because of parameter initialization, mini-batch ordering, and stochastic optimization. To reduce the variance associated with an individual training run, three 1D expert models are trained under identical data, architecture, and optimization settings using the random seeds 42, 123, and 2025. The final ensemble consists of three models trained on the complete training set, rather than the fold-specific checkpoints produced during cross-validation.

For sample j, the toxicity probability produced by the model trained with seed k is denoted by $p_j^{(k)} = \sigma(z_j^{(k)})$, $k \in \{1,2,3\}$, and the uncalibrated ensemble score is obtained by equal-weight arithmetic averaging:

$$s_j = \frac{1}{3}\sum_{k=1}^{3} p_j^{(k)}. \tag{8}$$

No individual seed is selected according to its performance on Test1 or Test2, and no seed-specific ensemble weight is introduced.

Although probability averaging reduces prediction variance, neural-network probabilities may remain systematically overconfident or underconfident. Platt scaling [23] is therefore applied after the three-model probability average. Training OOF predictions are first generated using the fixed group-aware five-fold partition; each training peptide receives an OOF probability from a model trained without the fold containing that peptide, and the OOF probabilities from the three random seeds are averaged to obtain the OOF ensemble score s_j^{OOF}. A two-parameter logistic calibrator is fitted to these OOF scores:

$$p_{1D,j} = \sigma\left(a\,\mathrm{logit}\left(\widetilde{s}_j\right) + b\right), \tag{9}$$

where $\mathrm{logit}(s) = \ln(s/(1-s))$, $\widetilde{s}_j = \mathrm{clip}(s_j, \varepsilon, 1-\varepsilon)$ with $\varepsilon = 10^{-6}$, and a and b denote the Platt calibration parameters; the clip prevents infinite logits for probabilities of exactly 0 or 1, and the same clipping is applied during fitting. The parameters are estimated by minimizing binary cross-entropy on the OOF scores, and the scale is reparameterized as $a = \mathrm{clip}(\exp(\eta),\, 0.02, 50) > 0$, which guarantees a monotonic transformation and therefore preserves the sample ranking of the ensemble score. For benchmark test or previously unseen peptides, the frozen calibrator is applied to the three-seed average: $p_{1\mathrm{D}} = \sigma\left(a\,\mathrm{logit}\left(\mathrm{clip}\left(\frac{1}{3}\sum_{k=1}^{3} p^{(k)},\ \varepsilon,\ 1-\varepsilon\right)\right) + b\right)$. The labels of Test1 and Test2 are not used for ensemble-member selection, calibration fitting, or 1D threshold optimization. The calibrated probability $p_{1\mathrm{D}}$ is the sequence-expert output used in

the top-level probability fusion described in Section 2.2; no additional Platt calibrator is fitted after the final GHXTox probability fusion.

2.4 pLDDT-Aware Multi-Scale 3D Structural Modeling

Peptide toxicity is influenced not only by amino acid composition and linear residue order but also by the three-dimensional (3D) arrangement of charged groups, hydrophobic regions, aromatic systems, and sulfur-containing moieties. However, peptides are generally short, conformationally flexible, and poorly covered by experimentally determined structures. Treating a single predicted conformation as an error-free structural observation may therefore amplify structure-prediction errors. Conversely, representing a peptide with only Cα coordinates or peptide-level summary descriptors fails to capture the local spatial-chemical environments that are relevant to toxicity.

To address these limitations, GHXTox employs a pLDDT-aware multi-scale structural expert that captures complementary information at five levels: residue semantics, spatial residue neighborhoods, full-atom chemical sites, peptide-level global properties, and conformer-ensemble statistics. The model first extracts residue-level Cα coordinates, full-atom coordinates, and normalized pLDDT confidence scores from the predicted structure. An internal sequence-semantic subbranch combines amino acid residue types, functional-group categories, explicit physicochemical descriptors, and ESM2 residue embeddings to generate contextualized sequence states. In parallel, a spatial-geometric subbranch constructs a hybrid residue graph containing both sequential adjacency edges and spatial-neighborhood edges and propagates geometric information through pLDDT-aware equivariant graph layers.

To overcome the limited chemical resolution of a Cα-only residue graph, GHXTox additionally extracts typed chemical sites from the full-atom structure and performs message passing using dual-scale distances, directional relationships, interaction-compatibility channels, and structural confidence. For peptides with available conformer ensembles, an additional statistical residual branch summarizes residue fluctuations, distance variation, and contact occupancy. The residual is constrained to exactly zero when conformer information is unavailable.

Rather than discarding low-confidence structures through hard filtering, GHXTox treats pLDDT as a continuous reliability signal and incorporates it into geometric edge weighting, chemical-site messages, residue-level structural gating, and peptide-level structural gating. Low-confidence structural evidence is therefore down-weighted rather than removed, preserving structural coverage while limiting the influence of uncertain coordinates. pLDDT is interpreted here as a local confidence estimate produced by the structure-prediction model, not as an experimentally determined measure of structural accuracy, conformational flexibility, or biological activity. The detailed architecture of the structural expert is illustrated in Figure 2.

Figure 2: Detailed architecture of the sequence-enhanced structural expert. The structural expert integrates ESM2-enhanced residue semantics, a pLDDT-aware geometric residue graph, full-atom chemical-site message passing, and availability-masked conformer-ensemble statistics. Residue-level structural states are continuously modulated by node-level confidence before serving as keys and values in sequence–structure cross-attention. A peptide-level confidence gate further controls the fused structural context, while conformer statistics provide a zero-initialized logit residual only when valid ensemble information is available.

2.4.1 Peptide Structure Prediction and Conformer Preparation

For a peptide sequence of length L, $S = (a_1, a_2, \ldots, a_L)$, ESMFold is first used to predict its three-dimensional structure [11]. The predicted structure is stored as a full-atom PDB file,

from which residue-level Cα coordinates, backbone coordinates, full-atom-derived chemical-site coordinates, and pLDDT scores are extracted. This operation is denoted

$$(\mathbf{X}, \mathcal{A}, \mathbf{q}) = \mathrm{ESMFold}(\mathcal{S}), \qquad \mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L\times 3}, \quad \mathbf{q} = (q_1, \ldots, q_L) \in [0,1]^L,$$

where $\mathbf{X}$ contains the Cα coordinates, $\mathcal{A}$ denotes the full-atom-PDB-derived backbone and chemical-site geometry, and $\mathbf{q}$ contains the residue-level pLDDT confidence scores rescaled to $[0, 1]$. The Cα coordinates serve as the primary spatial variables of the residue-level equivariant graph, whereas the full-atom coordinates are used to construct local chemical sites. Before geometric encoding, the coordinates are centered at the peptide centroid to remove the effect of absolute translation.

A single predicted structure provides a common structural anchor but cannot represent potential conformational heterogeneity. The conformer-residual stage was therefore trained on a multi-source conformer variant using a 2,795-peptide cohort with up to eight validated, aligned conformers per peptide. PepFlow supplies 761 training records of 11–15 residues, whereas AlphaFold2/ColabFold supplies 2,034 records of 16–30 residues [12–14]. These sources are represented explicitly in the residual head; nevertheless, generator identity remains partially coupled to peptide length and is treated as a limitation rather than as an independently identified causal factor.

For controlled ablation only, conformer statistics were admitted from a validated cache whose source and sequence-matching policies were fixed without using Test1 or Test2 labels. The Test1 cache contains eight conformers for each of 486 peptides: 130 PepFlow-derived records of 3–15 residues and 356 AlphaFold2/ColabFold-derived records of 16–30 residues. Test2 reuses 284 of these ensembles through an exact unique-sequence join. These caches are not used by the final benchmark predictor or the public inference interface. For every admitted ensemble, residue indices are matched to a common sequence order. An ensemble is represented as $\mathcal{C} = \{\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(M)}\}$, $1 \le M \le 8$, where $\mathbf{X}^{(m)} \in \mathbb{R}^{L\times 3}$ denotes the Cα coordinates of conformer m. Rigid-body alignment is performed before the ensemble statistics are calculated, ensuring that the resulting variation reflects internal geometric changes rather than arbitrary global rotations or translations.

Because conformer ensembles are not available for every training peptide, a binary availability indicator is defined as

$$c_{\mathrm{conf}} = \begin{cases} 1, & \text{if valid ensemble statistics are available,} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

When $c_{\mathrm{conf}} = 0$, the conformer-statistics tensors are filled with zeros, and the corresponding residual contribution is explicitly suppressed by the availability indicator. Missing ensemble information is therefore not interpreted as evidence of structural rigidity; instead, the model falls back exactly to the single-structure pathway. Under the benchmark ablation, valid conformer statistics are available for 486/1,126 (43.16%) Test1 peptides and 284/582 (48.80%) Test2 peptides. For the main comparison, the availability indicator is fixed to zero for every benchmark peptide and the structural probability is read from the single-structure base logits.

The generated conformers are computational samples rather than experimentally determined trajectories or rigorous thermodynamic ensembles. Consequently, RMSF, contact occupancy, and distance variance are interpreted as sampling-derived descriptors of conformational variation rather than direct measurements of peptide dynamics in solution.

2.4.2 Residue Feature Construction and Hybrid Geometric Graph

The structural expert constructs both an internal sequence representation and a spatial-geometric representation at the residue level. The internal sequence-semantic subbranch described here uses frozen ESM2 residue embeddings and a lightweight Transformer. It is distinct

from the LoRA-ProtT5 and bidirectional Mamba expert described in Section 2.3. The former supports residue-level sequence–structure alignment inside the structural expert, whereas the latter acts as an independent top-level 1D prediction expert.

For residue i, the input to the internal sequence-semantic subbranch is

$$\mathbf{u}_i^{\text{seq}} = \left[\mathbf{e}_i^{\text{AA}} \parallel \mathbf{e}_i^{\text{group}} \parallel \mathbf{f}_i^{\text{res}} \parallel P_{\text{ESM}}\left(\mathbf{e}_i^{\text{ESM2}}\right)\right], \tag{11}$$

where $\mathbf{e}_i^{\text{AA}} \in \mathbb{R}^{32}$ is the amino acid category embedding, $\mathbf{e}_i^{\text{group}} \in \mathbb{R}^{12}$ is the functional-group category embedding, $\mathbf{f}_i^{\text{res}} \in \mathbb{R}^{42}$ denotes the explicit residue-level physicochemical descriptors, and $\mathbf{e}_i^{\text{ESM2}} \in \mathbb{R}^{1280}$ is the ESM2-650M residue representation; the projection P_{ESM} maps the ESM2 representation to 128 dimensions. Stacking the residue-level feature vectors yields the sequence-feature matrix $U_{\text{seq}} \in \mathbb{R}^{L\times 214}$, where $214 = 32+12+42+128$. After a linear projection and positional encoding, a two-layer Transformer encoder with hidden dimension 128, four attention heads, and a feed-forward dimension of 512 produces the contextualized sequence states $\mathbf{H}^{\text{seq}} \in \mathbb{R}^{L\times 128}$. This representation encodes the contextual sequence semantics of each residue and subsequently serves as the query representation in the sequence–structure cross-attention module.

In the spatial subbranch, the initial representation of residue i is

$$\mathbf{h}_i^{(0)} = P_{\text{sp}}\left[\mathbf{f}_i^{\text{res}} \parallel \mathbf{g}_i^{\text{geo}} \parallel q_i \parallel P_{\text{ESM}}\left(\mathbf{e}_i^{\text{ESM2}}\right)\right],$$

where $\mathbf{g}_i^{\text{geo}} \in \mathbb{R}^{16}$ is a lightweight geometric descriptor derived from the Cα coordinates and pLDDT scores. The 16-dimensional descriptor includes (i) residue-level pLDDT and local confidence statistics; (ii) radial position, contact fractions, and soft spatial density; (iii) nearest-neighbor, nonlocal-neighbor, and local sequential distances; and (iv) sine/cosine encodings of the Cα-trace angles and torsions. These explicit descriptors provide stable local geometric priors without introducing an overly complex handcrafted representation for short and uncertain predicted peptide structures.

GHXTox then constructs a hybrid residue graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\text{seq}} \cup \mathcal{E}_{\text{knn}}), \tag{12}$$

where each residue corresponds to one node. Sequential edges $\mathcal{E}_{\text{seq}}$ connect adjacent residues and preserve peptide-chain continuity, whereas $\mathcal{E}_{\text{knn}}$ connects each residue to its 10 nearest spatial neighbors and is symmetrized. The hybrid graph therefore captures both covalent sequence order and nonlocal spatial contacts while avoiding large variations in graph density that may arise from a single fixed-distance cutoff. Equivalently, $A_{ij} \in \{0,1\}$ denotes whether residues i and j are connected by a sequential or k-nearest-neighbor edge in the hybrid graph.

For residues i and j, the Cα distance $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ is encoded by 16 radial basis functions with centers μ_k uniformly spaced over 0–24 Å, $\phi_k(d_{ij}) = \exp[-\gamma(d_{ij} - \mu_k)^2]$ with $\gamma = 1/(\mu_{k+1} - \mu_k)^2$. The base edge representation additionally includes the pairwise structural confidence and the normalized sequence separation:

$$c_{ij} = \sqrt{q_i q_j}, \qquad s_{ij} = \frac{\log(1+|i-j|)}{\log(L+1)}. \tag{13}$$

Thus, sequentially adjacent residues receive small values, whereas residues separated by longer sequence intervals approach one. The resulting edge feature is therefore $\mathbf{e}_{ij} = [\boldsymbol{\phi}(d_{ij}) \parallel c_{ij} \parallel s_{ij}]$. This formulation allows the model to distinguish sequentially adjacent contacts, long-range spatial contacts, and contacts supported by different levels of structural confidence.

2.4.3 pLDDT-Aware Equivariant Graph Encoding

To learn three-dimensional representations without relying on a particular global coordinate frame, the spatial branch employs three E(3)-equivariant graph neural network layers [24].

The network updates residue features using rotation- and translation-invariant quantities while updating coordinates equivariantly. Consequently, a global rigid transformation of the peptide structure does not change the predicted toxicity, whereas intermediate coordinate representations transform consistently with the input.

At layer ℓ, the message from residue j to residue i is computed as $\mathbf{m}_{ij}^{(\ell)} = \phi_e^{(\ell)}[\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell)}]$, where $\phi_e^{(\ell)}$ is a message multilayer perceptron. To prevent uncertain or distant contacts from contributing as strongly as local high-confidence contacts, a continuous edge weight is defined as

$$w_{ij}^{(\ell)} = A_{ij} \exp\left(-\frac{d_{ij}^{(\ell)}}{12}\right)\left[q_{\min} + (1 - q_{\min})\, c_{ij}^{\rho}\right], \qquad q_{\min} = 0.1, \quad \rho = 1.5, \tag{14}$$

The term $\exp(-d_{ij}^{(\ell)}/12)$ imposes a smooth distance decay with a characteristic decay length of 12 Å, and both the distance and the edge weight are recomputed at each layer from the current coordinates. Here, $A_{ij} \in \{0, 1\}$ is the adjacency indicator of the hybrid graph defined in Section 2.4.2, and c_{ij} is the pairwise structural confidence defined in Section 2.4.2. The minimum weight $q_{\min}$ prevents low-confidence regions from being completely disconnected, whereas the exponent ρ increases the relative contribution of high-confidence contacts. The distance-decay term further reduces the influence of more distant spatial neighbors.

Messages received by node i are aggregated with a normalization floor, and the EGNN layer updates node features and coordinates as

$$\begin{aligned}
\mathbf{m}_i^{(\ell)} &= \frac{\sum_j w_{ij}^{(\ell)}\, \mathbf{m}_{ij}^{(\ell)}}{\max\left(\sum_j w_{ij}^{(\ell)},\, 1\right)}, \\
\mathbf{h}_i^{(\ell+1)} &= \mathrm{LN}\left\{\mathbf{h}_i^{(\ell)} + \phi_h^{(\ell)}\left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)}\right]\right\}, \\
\mathbf{x}_i^{(\ell+1)} &= \mathbf{x}_i^{(\ell)} + 0.05\, \frac{\sum_j \alpha_{ij}^{(\ell)}\left(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\right)}{\max\left(\sum_j w_{ij}^{(\ell)},\, 1\right)}.
\end{aligned} \tag{15}$$

where $\alpha_{ij}^{(\ell)} = \tanh[\phi_x^{(\ell)}(\mathbf{m}_{ij}^{(\ell)})]\, w_{ij}^{(\ell)}$ is a bounded coordinate-update coefficient generated from the edge message and pairwise relative vectors. The coordinates are recentered after every layer to reduce numerical drift. Because the update depends only on scalar messages, distances, and relative coordinate vectors, the operation remains equivariant to rigid rotations and translations.

Three EGNN layers are stacked to progressively enlarge the spatial receptive field. The first layer primarily encodes direct geometric neighbors, whereas subsequent layers integrate higher-order spatial environments through repeated message passing. Unlike approaches that simply append pLDDT as an ordinary node feature, GHXTox uses pLDDT to directly modulate the strength of geometric messages, allowing structural uncertainty to influence representation learning at the propagation stage.

Continuous down-weighting does not imply that a low-pLDDT residue is biologically unimportant. Low confidence may arise from prediction error or potential conformational flexibility; the model reduces the reliability assigned to uncertain coordinate evidence while preserving the residue's sequence-semantic and physicochemical features.

2.4.4 Full-Atom Chemical-Site Message Passing

A Cα-based residue graph describes the global fold and spatial neighborhoods but cannot distinguish chemically distinct regions within the same side chain. For example, tyrosine contains both an aromatic ring and a hydroxyl group, whereas tryptophan contains an aromatic system and an indole nitrogen. Compressing each side chain into a single centroid would mix these

chemically different moieties. GHXTox therefore extracts chemistry-aware pseudo-sites from the predicted full-atom PDB structure and processes them with a dedicated local chemical branch.

Up to two chemical sites are kept for each residue. Site identity is represented by an eight-dimensional multi-hot vector $\mathbf{t}_{ir} \in \{0,1\}^8$ allowing overlapping types: positive, negative, donor, acceptor, aromatic, hydrophobic, sulfur, and conformational, where r indexes the chemical site within residue i. Multiple types may be assigned to the same site. For example, the cysteine sulfur site is labeled as donor, hydrophobic, and sulfur-containing, whereas the proline ring is labeled as hydrophobic and conformationally constrained. The aromatic ring and hydroxyl group of tyrosine are represented as separate sites, as are the aromatic system and the indole NE1 nitrogen (donor) of tryptophan.

A site coordinate is obtained from the mean position of the heavy atoms defining the corresponding chemical moiety. Site orientation is represented by either a bond direction or an unsigned plane normal. Because the sign of a plane normal is arbitrary, downstream directional features use absolute cosines, preserving rotation invariance and eliminating orientation-sign ambiguity.

The initial representation of site r in residue i is

$$\mathbf{z}_{ir}^{(0)} = \phi_{\text{site}}\left[\mathbf{t}_{ir} \parallel q_i \parallel o_{ir}^{\text{valid}}\right],$$

where q_i is the pLDDT score of the parent residue and o_{ir}^{valid} indicates whether a reliable orientation could be constructed; the input is projected into a 64-dimensional hidden space.

Chemical-site edges are established between valid sites separated by no more than 16 Å. Here, same-residue site edges are kept, preserving spatial relationships between chemically different moieties in complex side chains. Each edge contains four complementary feature families. First, the raw distance in Å is encoded using 16 radial basis functions, $\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \in \mathbb{R}^{16}$. Second, a peptide-size-normalized distance is constructed. The radius of gyration of the Cα coordinates is

$$R_g = \sqrt{\frac{1}{L}\sum_{i=1}^{L} \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \qquad \bar{\mathbf{x}} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i, \qquad \bar{d}_{uv} = \frac{d_{uv}}{R_g}, \tag{16}$$

where $\bar{\mathbf{x}}$ denotes the centroid of the valid Cα coordinates, and the normalized distance is encoded by eight radial basis functions over the interval 0–4, $\boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \in \mathbb{R}^8$. The raw representation describes the physical separation in Å, whereas the normalized representation describes the separation relative to the overall peptide size. Their combination allows the model to distinguish, for example, an 8 Å contact in a compact short peptide from an 8 Å contact in a larger peptide.

Third, four rotation-invariant orientation features are used:

$$\mathbf{o}_{uv} = \left[|\mathbf{o}_u \cdot \mathbf{r}_{uv}|,\ |\mathbf{o}_v \cdot \mathbf{r}_{vu}|,\ |\mathbf{o}_u \cdot \mathbf{o}_v|,\ o_{uv}^{\text{valid}}\right], \tag{17}$$

which describe the alignment of the source orientation with the intersite direction, the alignment of the target orientation with the reverse direction, the mutual orientation of the two sites, and orientation availability. Fourth, six explicit compatibility channels are used: opposite-charge, donor–acceptor, aromatic–aromatic, positive–aromatic (cation–π), sulfur–sulfur, and hydrophobic–hydrophobic compatibility. These channels encode candidate chemical compatibility patterns available to the model; they do not imply that a true salt bridge, hydrogen bond, cation–π interaction, disulfide bond, or hydrophobic core is present in the predicted structure.

The full chemical-site edge representation is

$$\mathbf{e}_{uv}^{\text{chem}} = \left[\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \parallel \boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \parallel \mathbf{o}_{uv} \parallel \mathbf{c}_{uv}^{\text{chem}} \parallel \sqrt{q_u q_v}\right]. \tag{18}$$

At layer l, each site message is computed from the two site states and the edge representation, confidence-weighted, and aggregated over the neighborhood. After two message-passing layers,

valid sites belonging to the same residue are mean-pooled, and the residue-level chemical residual and the chemical-enhanced spatial state are

$$\begin{aligned}
\mathbf{m}_{uv}^{(l)} &= \phi_{\text{chem}}^{(l)}\left[\mathbf{z}_u^{(l)} \parallel \mathbf{z}_v^{(l)} \parallel \mathbf{e}_{uv}^{\text{chem}}\right], \\
\bar{\mathbf{m}}_v^{(l)} &= \frac{\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v}\, \mathbf{m}_{uv}^{(l)}}{\max\left(\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v},\, 1\right)}, \\
\Delta\mathbf{h}_i^{\text{chem}} &= \mathbf{W}_{\text{chem}} \operatorname*{Mean}_{r\in\mathcal{S}_i}\left(\mathbf{z}_{ir}^{(2)}\right), \\
\mathbf{h}_i^{\text{sp}} &= \mathbf{h}_i^{\text{EGNN}} + \Delta\mathbf{h}_i^{\text{chem}},
\end{aligned} \tag{19}$$

where the projection $\mathbf{W}_{\text{chem}}$ and its bias $\mathbf{b}_{\text{chem}}$ are zero-initialized, so that $\Delta\mathbf{h}_i^{\text{chem}} = 0$ at the beginning of training. This conservative residual design allows the new chemical-site branch to progressively learn useful corrections from an intact geometric backbone rather than perturbing the existing spatial representation with a randomly initialized contribution.

2.4.5 Availability-Masked Conformer-Ensemble Residual

A single predicted structure provides a deterministic spatial anchor, whereas a peptide may adopt multiple related conformations. Running the complete geometric network independently on every conformer would be computationally expensive and could allow the model to exploit differences between conformer generators or uneven ensemble coverage. The current GHXTox model therefore uses a lightweight conformer-statistics residual. The single-structure expert remains the base predictor, and ensemble statistics are permitted to correct its logit only when valid conformer information is available.

For a peptide with M aligned conformers, we compute the residue-level root-mean-square fluctuation, the per-pair mean distance and distance standard deviation, and the contact occupancy at cutoff $r \in \{6, 8, 10\}$ Å:

$$\begin{aligned}
\text{RMSF}_i &= \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left\|\mathbf{x}_i^{(m)} - \bar{\mathbf{x}}_i\right\|_2^2}, \qquad \bar{d}_{ij} = \frac{1}{M}\sum_{m=1}^{M} d_{ij}^{(m)}, \\
\sigma_{ij}^{d} &= \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left(d_{ij}^{(m)} - \bar{d}_{ij}\right)^2}, \quad O_{ij}^{(r)} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\left(d_{ij}^{(m)} < r\right),
\end{aligned} \tag{20}$$

with $\bar{\mathbf{x}}_i$ the mean conformer coordinate of residue i. The 8 Å contact-occupancy uncertainty $U_{ij}^{(8)} = O_{ij}^{(8)}(1 - O_{ij}^{(8)})$ is also used: values close to zero or one indicate contacts that are consistently absent or present across the sampled conformers, whereas values near 0.5 indicate contact switching.

The ensemble statistics are organized into an eight-dimensional residue-level descriptor $\mathbf{r}_i^{\text{conf}} \in \mathbb{R}^8$, containing the residue RMSF; the mean pairwise distance to other residues; the standard deviation of the mean pairwise distances associated with the residue; the mean pairwise-distance standard deviation; the mean contact occupancies at 6, 8, and 10 Å; and the mean 8 Å contact-occupancy uncertainty, which together form the eight-dimensional descriptor. A twelve-dimensional peptide-level descriptor $\mathbf{g}^{\text{conf}} \in \mathbb{R}^{12}$ is also constructed, containing the normalized number of conformers; the mean, standard deviation, and maximum of RMSF; the mean, standard deviation, and maximum of the conformer RMSD; the mean and standard deviation of the radius of gyration; the mean and maximum of the pair-distance standard deviation; and the global mean of $U_{ij}^{(8)}$, which together form the twelve-dimensional descriptor. Before neural encoding, distance- and fluctuation-based statistics are rescaled by fixed constants to place heterogeneous descriptors on comparable numerical ranges; the exact scaling factors are reported in the implementation-details table.

The residue-level ensemble features are encoded and aggregated using both masked mean and masked maximum pooling, whereas the global descriptor is processed by a separate encoder, giving the ensemble embeddings $\mathbf{h}_{\text{conf}}^{\text{res}}$ and $\mathbf{h}_{\text{conf}}^{\text{global}}$. The generator is encoded by a three-dimensional one-hot vector $\mathbf{s}_{\text{conf}} \in \mathbb{R}^3$ for PepFlow, AlphaFold2/ColabFold, or another admitted generator. A learned source embedding $\phi_s(\mathbf{s}_{\text{conf}})$ is concatenated with the base structural embedding and the two ensemble embeddings to produce a raw logit residual

$$\Delta z_{\text{conf}}^{\text{raw}} = \phi_{\text{conf}}\left[\mathbf{h}_{\text{base}} \parallel \mathbf{h}_{\text{conf}}^{\text{res}} \parallel \mathbf{h}_{\text{conf}}^{\text{global}} \parallel \phi_s(\mathbf{s}_{\text{conf}})\right],$$

whose final linear layer is initialized to zero. Using the availability indicator c_{conf} defined in Eq. (10), the effective residual and the final structural-expert logit are

$$\begin{aligned} \lambda_{\text{conf}} &= \sigma\left(\beta_{\text{conf}}\right), \\ \Delta z_{\text{conf}} &= c_{\text{conf}}\, \lambda_{\text{conf}}\, \Delta z_{\text{conf}}^{\text{raw}}, \\ z_{\text{SE3D}} &= z_{\text{base}} + \Delta z_{\text{conf}}, \qquad c_{\text{conf}} = 0 \;\Rightarrow\; z_{\text{SE3D}} = z_{\text{base}}. \end{aligned} \tag{21}$$

where the effective conformer weight is initialized to 0.25 at the beginning of training through $\beta_{\text{conf}}^{(0)} = \text{logit}(0.25)$, so that $\sigma(\beta_{\text{conf}}^{(0)}) = 0.25$. Zero-filled feature tensors alone would not guarantee a zero correction because the residual network may contain nonzero biases; the additional hard availability mask ensures an exact fallback to the single-structure prediction. The conformer branch is permitted only to refine the single-structure predictor rather than replace the geometric backbone, reducing the risk of overfitting to limited and uneven ensemble coverage.

2.4.6 pLDDT-Aware Sequence–Structure Cross-Attention

After the EGNN and the chemical-site residual are applied, the model obtains the chemical-enhanced spatial representation

$$\mathbf{H}^{\text{sp}} = [\mathbf{h}_1^{\text{sp}}, \ldots, \mathbf{h}_L^{\text{sp}}]^\top \in \mathbb{R}^{L \times 128},$$

where $\mathbf{h}_i^{\text{sp}}$ is defined in Eq. (19). Spatial information alone does not determine whether a structural pattern is relevant to the current sequence context. For example, similar hydrophobic contacts may have different implications depending on residue identities and sequence positions. GHXTox therefore uses pLDDT-aware cross-attention within the structural expert, allowing contextualized sequence states to selectively retrieve spatial evidence.

A node-level gate is first constructed for each residue from the 42-dimensional residue descriptors and the pLDDT score:

$$\begin{aligned} g_i^{\text{base}} &= \phi_{\text{node}}\left[\mathbf{f}_i^{\text{res}} \parallel q_i\right], \\ \mathbf{c}_i^{\text{node}} &= \left[\frac{q_i - \mu_q}{T_q},\ \frac{(q_i - \mu_q)^2}{T_q^2},\ \mathbb{I}(q_i < 0.55)\right], \\ r_i^{\text{node}} &= 0.5 + \sigma\left(\mathbf{w}_n^\top \mathbf{c}_i^{\text{node}} + b_n\right), \\ g_i^{\text{node}} &= \text{clamp}\left(g_i^{\text{base}}\, r_i^{\text{node}},\ 0, 1\right), \end{aligned} \tag{22}$$

where $\mu_q = 0.70$, $T_q = 0.12$, and the residual confidence vector $\mathbf{c}_i^{\text{node}}$ increases the sensitivity of the gate to the distribution of structural confidence. The spatial representation is then continuously down-weighted as $\widehat{\mathbf{H}}^{\text{sp}} = \mathbf{H}^{\text{sp}} \odot \mathbf{g}^{\text{node}}$, with the node-level gate vector $\mathbf{g}^{\text{node}} \in [0,1]^L$ broadcast across the feature dimension.

In the cross-attention operation, the contextualized sequence representation serves as the query, and the gated spatial representation serves as the key and value:

$$\mathbf{H}^{\text{cross}} = \text{MHA}\left(Q = \mathbf{H}^{\text{seq}}, K = \widehat{\mathbf{H}}^{\text{sp}}, V = \widehat{\mathbf{H}}^{\text{sp}}\right). \tag{23}$$

This direction has a direct interpretation: each residue position retrieves spatial evidence according to its contextual sequence representation. The operation is not a direct feature interaction between the LoRA-ProtT5–Mamba expert and the structural expert; it occurs internally between the ESM2-enhanced sequence states and the geometric states of the structural expert.

In addition to residue-level gating, a peptide-level gate is generated from the overall structure-confidence profile. The model first summarizes the profile by the mean pLDDT $\bar{q}$, the minimum pLDDT $q_{\min}$, and the low-confidence fraction $r_{\text{low}} = \frac{1}{L}\sum_{i=1}^{L} \mathbb{I}(q_i < 0.55)$, and constructs the peptide-level confidence vector $\mathbf{c}^{\text{pep}} = [(\bar{q} - \mu_q)/T_q,\ (q_{\min} - \mu_q)/T_q,\ r_{\text{low}}]$. The base peptide-level gate and its residual-adjusted form are

$$\begin{aligned} g_{\text{pep}}^{\text{base}} &= \phi_{\text{global}}\left[\bar{q},\ q_{\min},\ r_{\text{low}}\right], \\ r_{\text{pep}} &= 0.5 + \sigma\left(\mathbf{w}_{\text{pep}}^{\top}\mathbf{c}^{\text{pep}} + b_{\text{pep}}\right), \\ g_{\text{pep}} &= \text{clamp}\left(g_{\text{pep}}^{\text{base}}\, r_{\text{pep}},\ 0, 1\right), \end{aligned} \tag{24}$$

where the peptide-level residual scale is analogous to, but separately parameterized from, the node-level residual scale, and $g_{\text{pep}} \in [0, 1]$ denotes the scalar peptide-level structural gate.

The final residue-level fusion representation combines the sequence state with the gated structural context, $\mathbf{H}^{\text{fused}} = \text{LN}\left[\mathbf{H}^{\text{seq}} \parallel g_{\text{pep}}\, \mathbf{H}^{\text{cross}}\right]$. The two gating levels answer complementary questions: how reliable the structural evidence is at an individual residue position, and how strongly the structural branch should be trusted for the peptide as a whole. The sequence representation is always kept, whereas the structural context is dynamically scaled according to prediction confidence. Consequently, when the predicted structure has low overall confidence, the structural expert can still rely on its internal sequence-semantic pathway without allowing uncertain coordinates to dominate the classification.

2.4.7 Global Feature Integration and Structural Classification

After pLDDT-aware cross-attention, each valid residue is represented by a 256-dimensional fused vector obtained by concatenating the 128-dimensional internal sequence state with the 128-dimensional structural context. Masked mean pooling is applied over the valid residue positions, and the pooled vector is concatenated with five global physicochemical descriptors before a three-layer MLP produces the base structural logit:

$$\begin{aligned} \mathbf{h}_{\text{pool}} &= \frac{\sum_{i=1}^{L} m_i\, \mathbf{H}_i^{\text{fused}}}{\sum_{i=1}^{L} m_i} \in \mathbb{R}^{256}, \\ \mathbf{h}_{\text{cls}} &= \left[\mathbf{h}_{\text{pool}} \parallel \mathbf{f}_{\text{pep}}^{\text{phys}}\right] \in \mathbb{R}^{261}, \\ z_{\text{base}} &= \phi_{\text{cls}}\left(\mathbf{h}_{\text{cls}}\right), \end{aligned} \tag{25}$$

where the mask excludes padding positions from the pooled representation, and $\mathbf{f}_{\text{pep}}^{\text{phys}} \in \mathbb{R}^5$ contains the normalized peptide length, the length-normalized net charge, the aromatic-residue fraction, the cysteine fraction, and the scaled mean hydropathy. ϕ_{cls} is a three-layer MLP with dimensions $261 \to 128 \to 64 \to 1$, each hidden layer applying GELU activation and dropout. Unlike maximum pooling, masked mean pooling integrates structural evidence distributed across residue positions, which is appropriate when toxicity-related spatial patterns are not confined to a single local region.

Applying the availability-masked conformer correction defined in Eq. (21) yields the final structural-expert logit, and the structural-expert probability is

$$z_{\text{SE3D}} = z_{\text{base}} + c_{\text{conf}}\, \lambda_{\text{conf}}\, \Delta z_{\text{conf}}^{\text{raw}}, \qquad p_{\text{SE3D}} = \sigma\left(z_{\text{SE3D}}\right),$$

as defined in Eq. (21). The resulting probability is passed directly to the dual-expert fusion in Section 2.2; the standalone structural threshold is used only for branch-specific analyses.

2.5 Training and Model-Selection Protocol

The two experts were trained independently because they differ in input representation, parameter scale, and optimization behavior, and they are combined only at the probability level. Group-aware training out-of-fold (OOF) predictions were used for calibration, fusion-weight selection, and threshold determination, and Test1/Test2 labels were excluded from all of these procedures. Highly similar peptides are confined to the same fold (Section 2.5.1); because both test sets were observed during development, they serve as frozen retrospective benchmarks rather than fresh external validation.

2.5.1 Group-Aware Five-Fold OOF Generation

The training set $\mathcal{D} = \{(\mathcal{S}_i, y_i, g_i)\}_{i=1}^{N}$, $N = 6387$, consists of peptide sequences $\mathcal{S}_i$, toxicity labels $y_i \in \{0, 1\}$, and sequence-homology groups g_i. A fixed five-fold manifest was constructed by CD-HIT clustering of the training sequences at a sequence-identity threshold of 0.80; this fold-level grouping is distinct from the 0.9 redundancy-removal threshold used for dataset construction in Section 2.1. The data are partitioned into five disjoint validation folds:

$$\begin{aligned} \mathcal{D} = \bigcup_{k=1}^{5} \mathcal{D}_k, \qquad & \mathcal{D}_k \cap \mathcal{D}_l = \varnothing, \quad k \neq l, \\ g_i = g_j \Longrightarrow f(i) = f(j), & \end{aligned} \tag{26}$$

where $f(i)$ denotes the fold assigned to sample i; all peptides in the same homology group therefore remain in the same fold. For fold k, the model is trained only on the complement $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$ and evaluated on the unseen fold $\mathcal{D}_k$. The OOF probability of sample i is defined as

$$\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k, \qquad \hat{p}_i^{\mathrm{OOF}} = F_{\boldsymbol{\theta}^{(-k)}}\left(\mathcal{S}_i\right), \quad i \in \mathcal{D}_k, \tag{27}$$

so each training peptide receives exactly one probability from a model that did not observe it during optimization.

For the 1D expert, seed-specific OOF probability vectors were aligned and averaged before calibration (Section 2.3.4). The structural expert uses the same folds; each fold-specific checkpoint produces OOF probabilities for its held-out fold, with conformer statistics handled by the availability mask defined in Section 2.4.5. Parameters selected from the complete OOF table remain development-stage choices and are evaluated on independent test sets.

2.5.2 Loss Functions and Class-Imbalance Handling

The 1D expert uses the positive-class-weighted binary cross-entropy defined in Section 2.3.3, with class counts recomputed from each fold's training partition; no synthetic or point-mutated samples were used.

The structural expert is trained in stages. In the base geometric and chemical-site stage, the full-atom chemical-site branch is initialized from a trained pLDDT-aware geometric model while the remaining backbone is frozen, using the same weighted BCE. During this stage, pLDDT-dependent Gaussian coordinate perturbations are applied:

$$\sigma_i = \sigma_0 \left[1 + \kappa \left(1 - q_i\right)\right], \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \sigma_i^2 \mathbf{I}_3\right), \qquad \widetilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\varepsilon}_i, \tag{28}$$

with $\sigma_0 = 0.12$ Å and $\kappa = 2.0$. The same displacement is applied to the backbone, functional-group, and chemical-site coordinates so that intra-residue geometry is preserved, and low-pLDDT residues receive stronger perturbations than high-confidence residues; no perturbation is used during validation or inference.

In the conformer-ensemble residual stage, the pLDDT-aware geometric backbone, internal sequence encoder, chemical-site branch, and base classifier are frozen, and only the conformer

encoders and residual MLP are optimized. This stage uses unweighted BCE with a residual-magnitude penalty:

$$\mathcal{L}_{\text{conf}} = \text{BCEWithLogits}\left(z_{\text{SE3D}}, y\right) + \lambda_{\Delta} \frac{\sum_{i=1}^{B} c_i \left|\Delta z_{\text{conf},i}\right|}{\max\left(\sum_{i=1}^{B} c_i,\, 1\right)}, \qquad \lambda_{\Delta} = 0.005, \tag{29}$$

where the penalty is computed only on samples with available conformer information. The residual layer is zero-initialized and gated by the hard conformer-availability mask, so the initial model is equivalent to the single-structure expert and uncovered samples receive exactly zero correction.

2.5.3 Optimization, Checkpoint Selection, and Hyperparameter Settings

Both experts were optimized with AdamW [25]; the principal hyperparameters and checkpoint criteria are summarized in Table 2. Random-number generators were seeded, and data were shuffled only during training.

Table 2: Main optimization and model-selection settings of the two GHXTox experts.

Module or stage	Batch size	Learning rate	Weight decay	Epochs	Checkpoint rule
1D LoRA-ProtT5 + Bi-Mamba	16	3×10^{-4}	1×10^{-2}	Fixed 15	No early stopping
3D chemical-site refinement	32	5×10^{-4}	1×10^{-4}	Max. 30	Best validation MCC; patience 8
Conformer residual refinement	32	1×10^{-3}	5×10^{-4}	Max. 50	Minimum validation loss; patience 8

The 1D expert was trained for a fixed 15 epochs without early stopping, and the final sequence probability is the average of three full-data models (seeds 42, 123, and 2025) rather than a seed selected by validation performance; this configuration yields approximately 13.1 million trainable parameters. The 3D chemical-site stage was initialized from a trained pLDDT-aware geometric checkpoint with the backbone frozen, and its checkpoint was selected by validation MCC with a patience of 8 epochs. The conformer-residual stage kept the geometric and chemical backbone frozen and selected its checkpoint by validation loss with the same patience. Gradients were clipped to a maximum norm of 5.0 during structural training.

For full-data refits, the epoch count was fixed from the median of the fold-specific selected epochs (8 for the single-structure chemical-site backbone and 6 for the multi-source conformer residual) and was never chosen using Test1 or Test2. In the final benchmark and public inference protocol, the conformer availability mask is fixed to zero and the structural probability is read from the single-structure base logits.

2.5.4 OOF-Based Calibration, Fusion-Weight Selection, and Threshold Freezing

After the expert-level OOF predictions were generated, model selection proceeded at the probability level: no network parameters were updated, and only training OOF probabilities were used. The three seed-specific 1D OOF vectors were averaged and calibrated with the monotonic Platt mapping defined in Section 2.3.4; the structural-expert probability was used directly without an additional top-level calibrator. The calibrated 1D probability and the structural probability were then combined as in Section 2.2, where $p_{i,\text{SE3D}}^{\text{OOF}}$ is the pure structure probability and contains no contribution from the 1D predictor. No stacking classifier or sample-specific routing network was fitted.

The fusion weight w and decision threshold τ were selected by maximizing MCC on the complete training OOF table, with BACC as the secondary criterion for ties:

$$(w^*, \tau^*) = \arg\max_{w,\tau} \ \mathrm{MCC}\left(\mathbf{y}, \hat{\mathbf{y}}(w, \tau)\right). \tag{30}$$

This yielded $w^*_{\mathrm{1D}} = 0.72$, $w^*_{\mathrm{SE3D}} = 0.28$, and $\tau^* = 0.47022256531284556$ (0.4702 in tables). The fused probability and class prediction follow the rules in Sections 2.2 and 2.6. A leave-one-fold-out check indicated that the selected parameters were not disproportionately dependent on any single fold. These values were then frozen and applied unchanged to Test1 and Test2; benchmark labels were not used for recalibration or parameter adjustment.

2.6 Evaluation Metrics

Because the benchmark data are class-imbalanced (Test2 contains 46 toxic and 536 non-toxic peptides), MCC and BACC were used as the primary threshold-dependent metrics, whereas ACC, sensitivity (SN), specificity (SP), precision, and F1 were reported for completeness. AUROC and AUPRC quantified ranking performance, and the Brier score and ECE-10 assessed probability quality. All threshold-dependent metrics were computed at the fixed decision threshold $\tau^* = 0.47022256531284556$ (rounded to 0.4702 in tables), determined from the complete training OOF table (Section 2.5.4) and applied unchanged to Test1 and Test2; ranking and calibration metrics were computed from the continuous probabilities p_i.

At the fixed threshold τ, binary predictions are $\hat{y}_i = \mathbb{I}[p_i \geq \tau]$ for N peptides with labels $y_i \in \{0, 1\}$ and predicted probabilities p_i; they are summarized by true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), with $\mathrm{ACC} = (\mathrm{TP} + \mathrm{TN})/N$, $\mathrm{SN} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, $\mathrm{SP} = \mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$, $F_1 = 2\mathrm{TP}/(2\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$, and $\mathrm{BACC} = (\mathrm{SN} + \mathrm{SP})/2$.

Matthews correlation coefficient.

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}, \tag{31}$$

with $-1 \leq \mathrm{MCC} \leq 1$. Because MCC jointly accounts for all four counts, it was used as the primary fixed-threshold summary metric and as the objective for OOF fusion-parameter selection [26].

Brier score and ranking metrics. AUROC is the probability that a randomly selected toxic peptide is ranked above a randomly selected non-toxic peptide; AUPRC was computed as average precision, whose chance-level baseline depends on positive-class prevalence. The Brier score is the mean squared difference between the predicted probability and the binary ground-truth label [27]:

$$\mathrm{Brier} = \frac{1}{N} \sum_{i=1}^{N} \left(p_i - y_i\right)^2. \tag{32}$$

It ranges from 0 to 1 and penalizes both incorrect predictions and excessive confidence.

Ten-bin expected calibration error. ECE-10 divides predictions into ten equal-width bins over $[0, 1]$, with the final bin including $p_i = 1$. Let $\mathcal{B}_m$ denote the m-th bin, with mean confidence $\mathrm{conf}(\mathcal{B}_m)$ and observed toxic-peptide frequency $\mathrm{freq}(\mathcal{B}_m)$:

$$\mathrm{ECE}_{10} = \sum_{m=1}^{10} \frac{|\mathcal{B}_m|}{N} \ \left|\mathrm{conf}\left(\mathcal{B}_m\right) - \mathrm{freq}\left(\mathcal{B}_m\right)\right|. \tag{33}$$

Values closer to zero indicate better agreement between predicted confidence and observed toxicity frequency. Because ECE depends on the binning scheme, the number and boundaries of the bins remain fixed across models and datasets.

2.7 Robustness, Uncertainty, and Statistical Analysis

A single point estimate cannot characterize the reliability of a deep predictor: performance varies with random initialization and with the finite evaluation sample, and aggregate metrics may conceal heterogeneity across subgroups. All analyses below were therefore performed on the frozen predictor, without retraining, recalibration, or threshold adjustment in response to their results. Detailed resampling procedures, seed-level results, and the training-OOF conformer information-ceiling diagnostics are reported in the Supporting Information.

2.7.1 Multi-Seed Reproducibility and Prediction Dispersion

To assess sensitivity to stochastic optimization, three full-data 1D models were trained with seeds 42, 123, and 2025 under identical data partitions, architecture, training duration, and hyperparameters. The 1D expert is the equal-weight probability average of the three calibrated models rather than the best-performing seed selected post hoc; a multi-checkpoint 3D ensemble was evaluated but did not satisfy the prespecified improvement criterion and was not adopted. For an evaluation metric M, seed-specific results may be summarized as $\bar{M} \pm \mathrm{SD}(M)$ across the three seeds, always with the same threshold.

Prediction dispersion is additionally quantified at the sample level: $u_{i,\mathrm{seed}}$ is the standard deviation of the three uncalibrated seed probabilities for peptide i. A large value flags predictions that are sensitive to optimization randomness and may mark boundary or expert-disagreement samples, but it is an empirical training-variability measure, not a Bayesian uncertainty estimate. Because the calibrated ensemble is a deterministic function of the three trained models, seed-level variability is summarized through $u_{i,\mathrm{seed}}$ rather than assigned to the averaged prediction.

2.7.2 Class-Stratified Bootstrap Confidence Intervals

Finite test sets are themselves samples, and this matters especially for Test2, where a small number of positive-class changes can substantially alter SN, BACC, and MCC. We therefore report 95% percentile confidence intervals from class-stratified bootstrap resampling that preserves the original size of each class [28]; the interval is formed from the empirical 2.5th and 97.5th percentiles of the replicate metric values. The primary test-set analysis used $B = 5000$ replicates, and subgroup-difference analyses used 1000. Within every replicate, threshold-dependent metrics retain the fixed decision threshold, which is never reoptimized on resampled data; AUROC, AUPRC, and the Brier score are recomputed from the resampled continuous probabilities. These intervals quantify finite-sample uncertainty conditional on the frozen predictor and the observed evaluation distribution; they do not incorporate training-seed variability or distribution shift and are therefore interpreted separately from the multi-seed and subgroup analyses.

2.7.3 Paired Bootstrap and Randomization Significance Testing

Comparisons between the current model and an internal baseline are paired because both models score the same peptides. We therefore analyze the metric difference $\Delta_M^{\mathrm{obs}} = M_B - M_A$ directly on aligned samples rather than comparing independent intervals. A paired class-stratified bootstrap applies the same resampled indices to both models, and its 95% percentile interval characterizes the difference; a within-sample exchange randomization test additionally evaluates whether the two model outputs are exchangeable for each peptide [29]. Both procedures used 5,000 replicates with a fixed random seed. For higher-is-better metrics, an interval entirely above zero supports a consistent positive difference; for lower-is-better measures such as the Brier score, improvement corresponds to an interval entirely below zero.

MCC and BACC were the prespecified primary summary measures; SN, SP, AUROC, AUPRC, and the Brier score provided complementary evidence, and their p-values are interpreted descriptively because several correlated metrics are examined together. Finally, because

Test1 and Test2 were observed during historical development, paired intervals and randomization tests on these sets provide retrospective evidence for model differences rather than prospective confirmation. Detailed exchange mechanics and the randomization p-value calculation are given in the Supporting Information.

2.7.4 Subgroup and Structure-Quality Analysis

Aggregate metrics may conceal heterogeneity across peptide lengths, physicochemical properties, and structure-quality ranges. We therefore conducted prespecified subgroup analyses without modifying the model or threshold. Length strata were 11–15, 16–30, and 31–50 residues; structural diagnostics additionally stratified by mean pLDDT ($<$0.55, 0.55–0.70, $\geq$0.70), net charge, mean hydropathy, cysteine count, and chemical-site coverage ($<$0.80, 0.80–0.95, $\geq$0.95), where site coverage is the fraction of residues containing at least one valid chemical site. A stratum was reported only when it contained at least 20 peptides and at least five from each class; ineligible strata were recorded but excluded from metric interpretation.

All subgroup metrics used the global threshold and the original continuous probabilities, with no stratum-specific threshold or model fitting. When the current model was compared with a baseline, the subgroup difference $\Delta M_g = M_{B,g} - M_{A,g}$ was accompanied by 1,000-replicate class-stratified bootstrap intervals for MCC and AUPRC. Because many variables and bins are examined, subgroup findings were treated as heterogeneity diagnostics and interpreted together with stratum sizes, neighboring-stratum trends, and interval overlap rather than as independent discoveries. pLDDT is a prediction-confidence estimate rather than experimental structural accuracy, so high-pLDDT performance does not by itself establish that the corresponding structures are experimentally correct.

2.7.5 Selective Prediction with an OOF-Derived Cutoff

Selective prediction permits abstention on high-uncertainty peptides while leaving the checkpoints, calibrated probabilities, fusion weights, binary threshold, and binary predictions unchanged. No auxiliary error model was fitted: for peptide i, uncertainty is the normalized proximity of the fused probability p_i to the fixed threshold τ:

$$d_i = 1 - \frac{|p_i - \tau|}{\max(\tau, 1-\tau)}, \qquad \tau = 0.47022256531284556. \tag{34}$$

A larger d_i indicates a prediction closer to the decision boundary and is therefore treated as less reliable. Expert disagreement, seed dispersion, pLDDT, and conformer-derived variables were not included in the formal policy.

The abstention cutoff was determined exclusively from group-aware training OOF predictions. A prespecified target coverage of 90% produced $d^*_{\mathrm{OOF}} = 0.47361765874159856$, after which the rule was fixed:

$$d_i \leq d^*_{\mathrm{OOF}} \Rightarrow \text{Accept}, \qquad d_i > d^*_{\mathrm{OOF}} \Rightarrow \text{Abstain}. \tag{35}$$

The margin-only score was selected from prespecified candidate uncertainty representations using training-OOF risk–coverage performance; once selected, its score computation and the benchmark accept/abstain decisions require neither class labels nor benchmark identity. No benchmark-specific quantile or target coverage was recomputed during transfer.

Risk–coverage behavior was summarized at OOF coverages of 100%, 95%, 90%, 85%, and 80% using accepted-sample error, MCC, BACC, sensitivity, and specificity. Equal-count random rejection, evaluated with 5,000 random masks that retained exactly the same number of observations as the frozen rule, served as the control, and paired class-stratified bootstrap intervals were calculated from 5,000 replicates while the frozen acceptance mask was indexed directly onto each resampled set.

2.8 Interpretability Analysis Protocol

Interpretability analyses are defined on the model and do not participate in feature selection, threshold selection, or fusion-weight optimization. Unless otherwise stated, probability changes are computed with all non-target inputs held fixed, summarized separately for toxic and non-toxic peptides, and accompanied by peptide-level bootstrap intervals. The analyses therefore describe model sensitivity and expert complementarity; they are not interpreted as evidence of a causal toxicophore.

2.8.1 Fixed-Representation Residue Occlusion and Seed Stability

Residue-level sensitivity was evaluated on a prespecified 16-peptide cohort selected before inspection of the attribution values: within each test set and each expert-outcome group, one toxic and one non-toxic peptide nearest the fusion threshold were chosen, with ties resolved by source index. For each of the three 1D checkpoints, one contextual residue representation was zeroed while all other representations and model parameters were kept fixed. The per-seed occlusion effect was the unoccluded minus occluded logit; the three raw seed probabilities were averaged before the frozen OOF-fitted Platt transform, and the displayed attribution was the calibrated probability change normalized by the within-peptide L_1 norm. Stability was summarized by residue-wise sign agreement and pairwise Spearman correlation of absolute effects across seeds. This experiment quantifies sensitivity to contextual residue vectors; it is not a token-mutation or Integrated Gradients analysis and does not test biological causality.

2.8.2 Chemical-Site Deletion Analysis

In an earlier secondary sensitivity analysis performed before test-time conformer activation was discontinued, each valid modeled chemical site was ablated by setting its availability mask, type vector, coordinate, orientation, and orientation-validity entry to zero. All non-target sequence features, residue geometry, pLDDT values, other chemical sites, conformer descriptors, model weights, and operating thresholds were held fixed. The complete frozen 3D forward pass was recomputed, including the conformer residual when a cache entry was available. Complementary perturbations deleted all sites carrying one chemical category or all site slots assigned to one residue. We report $\Delta p_{3D} = p_{3D}^{\text{base}} - p_{3D}^{\text{deleted}}$ and, with the 1D probability fixed, the induced fusion change $\Delta p_{\text{fused}} = 0.28\Delta p_{3D}$. Because the modeled sites and candidate spatial relations are derived from predicted structures, these quantities describe model sensitivity rather than confirmed chemical bonds or intervention effects. They are not used to claim an attribution gain for the final single-structure benchmark predictor.

2.8.3 pLDDT–Gate Association Analysis

The unperturbed structural expert was additionally run on all 1,126 Test1 and 582 Test2 peptides. For each peptide, we recorded the mean pLDDT, learned global gate, and the within-peptide Spearman correlation between residue-level pLDDT and node gates. Associations were summarized separately by test set and for the pooled descriptive sample. Ninety-five percent bootstrap percentile intervals were obtained from 2,000 peptide-level resamples with seed 2026. This analysis characterizes the association between predicted-structure confidence and learned gate values; it does not establish structural accuracy, feature attribution, or biological causality.

2.8.4 Expert Agreement and Conflict Analysis

To characterize expert complementarity, each Test1 and Test2 peptide was assigned to one of four mutually exclusive groups: both experts correct, only the 1D expert correct, only the 3D expert correct, or both incorrect. Standalone outcomes used the training-derived thresholds of

0.392 (1D) and 0.677819 (3D), whereas the fused predictor retained its separately fixed threshold. Using 1D errors as the reference, we computed the fraction rescued by the 3D expert overall and separately for toxic and non-toxic peptides, and compared the discordant groups with respect to peptide length, cysteine count, mean pLDDT, and chemical-site coverage. Group differences were summarized by Cliff's delta with 5,000 label-stratified bootstrap replicates and permutation tests with 10,000 replicates, with one Benjamini–Hochberg correction applied across the prespecified comparison family. Test1 and Test2 were analyzed separately. All analyses were performed on fixed per-sample predictions.

3 RESULTS AND DISCUSSION

3.1 Overall Predictive Performance on the Benchmark Test Sets

Using the OOF-frozen weight and threshold without dataset-specific retuning, GHXTox correctly classified 1,067 of 1,126 Test1 peptides (TP/TN/FP/FN = 287/780/26/33). This corresponds to ACC 0.9476, BACC 0.9323, SN 0.8969, SP 0.9677, F1 0.9068, MCC 0.8705, AUROC 0.9811, and AUPRC 0.9674. On the strongly imbalanced Test2 set, GHXTox correctly classified 551 of 582 peptides (35/516/20/11), yielding ACC 0.9467, BACC 0.8618, SN 0.7609, SP 0.9627, F1 0.6931, MCC 0.6673, AUROC 0.9433, and AUPRC 0.8046. The lower Test2 BACC and F1, despite high ACC, reflect the difficulty of recovering the 46 toxic peptides and demonstrate why class-balanced and precision–recall metrics are essential for this benchmark.

3.1.1 Comparison with Existing Peptide-Toxicity Predictors

Table 3 compares models reported on the same benchmark partitions. Values for ToxMSRC and ToxPLTC are transcribed from the corresponding publications or their accompanying result tables [2, 3]; NR denotes an unreported metric. GHXTox obtains the highest Test1 ACC, BACC, SP, MCC, AUROC, and AUPRC among the compared rows, while ToxPLTC has higher Test1 SN. On Test2, GHXTox obtains the highest ACC, SP, MCC, and reported AUPRC, but its BACC and SN remain below both reference models. The results therefore point to a stronger overall balance of specificity, correlation, and positive-class ranking, not comprehensive superiority on every metric. Because Test1 and Test2 had been observed historically, these differences remain retrospective benchmark evidence and require confirmation on a new prefrozen external set.

Table 3: Performance comparison on the benchmark Test1 and Test2 sets.

Dataset	Model	ACC	BACC	SN	SP	MCC	AUROC	AUPRC
Test1	ToxMSRC	0.9405	0.9217	0.8781	0.9653	0.8520	0.9650	NR
	ToxPLTC	0.9378	0.9302	**0.9125**	0.9479	0.8496	0.9709	0.9426
	GHXTox (current)	**0.9476**	**0.9323**	0.8969	**0.9677**	**0.8705**	**0.9811**	**0.9674**
Test2	ToxMSRC	0.9416	0.8689	0.7826	0.9552	0.6550	0.9430	NR
	ToxPLTC	0.9261	**0.8804**	**0.8261**	0.9347	0.6197	0.9295	0.7353
	GHXTox (current)	**0.9467**	0.8618	0.7609	**0.9627**	**0.6673**	**0.9433**	**0.8046**

3.1.2 Discrimination, Calibration, and Decision Characteristics

The final fused probabilities have Brier scores of 0.0410 and 0.0436 on Test1 and Test2, respectively. These values summarize mean squared probability error under the frozen inference protocol. ECE-10 is not reported here because a consistently regenerated value is not available for both current single-structure prediction files. Test2 AUPRC (0.8046) directly reflects the precision–recall trade-off for the rare toxic class, which AUROC does not capture. The 11 false negatives on Test2 also delineate the limits of the operating point; the model's high specificity should not be interpreted as complete minority-class coverage.

Figure 3: Overall discrimination and calibration performance of GHXTox. (A, B) Test1 ROC and PR curves; (C, D) Test2 ROC and PR curves; (E, F) Test1 and Test2 reliability diagrams.

3.2 Complementarity of the 1D and 3D Experts

The two experts are fused only after independent probability generation. The 1D branch provides the stronger standalone ranking signal, whereas the 3D branch contributes confidence-weighted geometry and local chemical context. This separation makes the gain attributable to complementary probability evidence rather than duplicated use of an earlier fused prediction.

3.2.1 Performance of the Individual Experts

Table 4 reports threshold-free discrimination and probability error for probability streams that are aligned with the final model. Platt scaling preserves the 1D ranking but reduces its Brier score. Relative to calibrated 1D alone, final fusion improves Test1 AUROC/AUPRC from 0.9799/0.9644 to 0.9811/0.9674 and reduces Brier from 0.0429 to 0.0410. On Test2, the corresponding values change from 0.9409/0.7831/0.0451 to 0.9433/0.8046/0.0436. The current fused predictor therefore adds complementary structural ranking information while maintaining a lower probability error than the calibrated 1D stream. The structural probability in these rows is the multi-source-trained checkpoint's single-structure base output; no benchmark conformer cache is activated.

Table 4: Performance of the 1D probability streams and the current probability-level fusion. The final row for each benchmark uses the single-structure base output of the multi-source-trained structural checkpoint.

Dataset	Probability stream	AUROC	AUPRC	Brier
Test1	1D, uncalibrated	0.9799	0.9644	0.0445
	1D, OOF Platt calibrated	0.9799	0.9644	0.0429
	Current fusion (0.72/0.28)	**0.9811**	**0.9674**	**0.0410**
Test2	1D, uncalibrated	0.9409	0.7831	0.0491
	1D, OOF Platt calibrated	0.9409	0.7831	0.0451
	Current fusion (0.72/0.28)	**0.9433**	**0.8046**	**0.0436**

3.2.2 Probability Fusion of the 1D and 3D Experts

The 0.72 sequence weight is an empirical result of the training-OOF search, not a manually imposed assertion that sequence information is intrinsically more important. Both expert weights and the full-precision decision threshold of 0.47022256531284556 were fixed before Test1 or Test2 evaluation. The held-out benchmark results therefore assess one prespecified operating point; no benchmark-specific reweighting, threshold adjustment, or post hoc selection between equal and unequal fusion is performed.

3.2.3 Expert Agreement, Disagreement, and Error Correction

The experts agreed on most benchmark peptides, but the discordant samples revealed a reproducible pattern of complementary error correction. On Test1, the four outcome groups contained 1,032 both-correct, 29 1D-only-correct, 24 3D-only-correct, and 41 both-wrong peptides. The corresponding Test2 counts were 529, 13, 15, and 25. Among the 65 Test1 peptides misclassified by the 1D expert, the 3D expert correctly classified 24, giving a rescue rate of 36.92% (95% Wilson CI, 26.23–49.08%). It rescued 15 of 40 1D errors on Test2, corresponding to 37.50% (24.22–52.97%). The structural pathway therefore did not merely reproduce the sequence expert's decisions; it supplied correct counter-evidence for more than one third of its errors on each benchmark.

The rescue pattern was strongly class-asymmetric. Nineteen of the 24 3D-only-correct Test1 peptides and 14 of the 15 corresponding Test2 peptides were non-toxic. Equivalently, the 3D expert rescued 19 of 37 non-toxic 1D errors on Test1 and 14 of 29 on Test2, compared with 5 of 28 and 1 of 11 toxic 1D errors, respectively. The structural expert mainly corrected false-positive predictions from the sequence expert, whereas improvements in toxic-peptide recall were limited.

Peptides classified correctly only by the 3D expert also tended to be shorter. Their median lengths were 20 versus 25 residues on Test1 and 18 versus 28 on Test2 when compared with the 1D-only-correct groups. The corresponding Cliff's deltas were -0.286 and -0.523. However, none of the ten prespecified split-by-attribute comparisons remained significant after Benjamini–Hochberg correction; the smallest adjusted value was $q = 0.156$ for Test2 peptide length. The length pattern is therefore descriptive rather than evidence that the structural expert is generally superior for short peptides. Cysteine count, mean pLDDT, and chemical-site coverage likewise showed no stable cross-benchmark separation. Conformer availability was not assigned an independent interpretation because of its dependence on length and generator coverage.

These results support a complementary rather than substitutive role for the structural expert. Figure 4 summarizes the outcome composition, conditional rescue rates, and prespecified discordant-group contrasts.

Figure 4: The 3D structural expert provides class-asymmetric corrections for a subset of errors made by the 1D sequence expert. (A) Distribution of Test1 and Test2 peptides among the both-correct, 1D-only-correct, 3D-only-correct, and both-wrong groups, defined using the standalone expert thresholds. (B) Conditional rescue rate of the 3D expert among all 1D errors and among toxic and non-toxic 1D errors. Structural rescue is concentrated primarily among non-toxic peptides on both benchmarks. (C) Prespecified effect-size estimates comparing 3D-only-correct with 1D-only-correct peptides for length, cysteine count, mean pLDDT, and chemical-site coverage; error bars denote 95% bootstrap intervals. None of the prespecified comparisons remained significant after Benjamini–Hochberg correction. Conformer availability is not shown as an independent explanatory variable because it is confounded with peptide length and structure-generator assignment.

3.3 Analysis of the Task-Adaptive 1D Sequence Expert

To investigate the role of the task-adaptive 1D expert, we first compared the current sequence architecture with the earlier sequence pipeline based on pooled ProtT5 representations and a TextCNN classifier [3], and further examined task adaptation, bidirectional state-space modeling, multi-seed ensembling, and OOF-based probability calibration. Unlike approaches that use a pretrained protein language model merely as a fixed feature extractor, GHXTox preserves residue-level ProtT5 representations, adapts them to peptide toxicity prediction through LoRA, and subsequently models residue dependencies in both the N-to-C and C-to-N directions using bidirectional Mamba.

The gain of the new 1D expert did not appear uniformly across threshold-dependent metrics. At their respective operating points, the previous ProtT5-TextCNN sequence model achieved BACC, MCC, AUROC, and AUPRC values of 0.9317, 0.8586, 0.9724, and 0.9436, respectively, whereas the LoRA-ProtT5–Bi-Mamba expert achieved 0.9300, 0.8562, 0.9761, and 0.9533. Thus, BACC and MCC remained broadly comparable, while AUROC and AUPRC increased by approximately 0.0037 and 0.0097, respectively.

The improvement was concentrated in probability ranking. This stronger ranking stream benefits subsequent fusion with structural evidence and explains why sequence-model comparisons should not rely on ACC or BACC alone.

3.3.1 Task Adaptation of ProtT5 and the Final 1D Representation

Existing peptide-toxicity predictors commonly employ protein language models to generate fixed sequence representations. For example, StrucToxNet uses ProtT5-derived sequence features within its graph representation, whereas PeptiTox integrates ESM2 embeddings with geometry derived from predicted structures [6, 7]. A major distinction of GHXTox is that ProtT5 is adapted to the toxicity task through LoRA, while its residue-level representations are preserved for downstream sequence modeling.

This design serves two complementary purposes. First, low-rank updates allow the pretrained semantic space to adapt to peptide toxicity prediction without requiring full fine-tuning of the ProtT5 backbone. Second, peptide-level pooling is deliberately postponed: contextualized residue states are passed to the Bi-Mamba encoder, preserving position-dependent information and allowing additional modeling of local and long-range residue dependencies.

Because LoRA and Bi-Mamba were not ablated independently, their contribution is evaluated jointly.

3.3.2 Bidirectional State-Space Modeling of Peptide Sequences

After task-adaptive ProtT5 encoding, GHXTox employs bidirectional Mamba rather than TextCNN or immediate peptide-level pooling for secondary sequence modeling. This choice is motivated by the properties of peptides: toxicity-related evidence may not be confined to a single local motif; instead, combinations of charged, hydrophobic, aromatic, or other residues may interact over distant sequence positions. Fixed-size convolutions can capture such dependencies only through progressively enlarged receptive fields, whereas a state-space model propagates contextual information through recurrent state updates.

Although peptide sequences possess a natural N-to-C direction, toxicity classification is not an autoregressive prediction task. GHXTox therefore performs both forward and backward selective scans and integrates their representations. Each residue can consequently be contextualized by information from both termini rather than forcing the classifier to adopt a single direction of information propagation.

Because no strict single-variable comparison between unidirectional and bidirectional Mamba under otherwise identical conditions is available, we do not assign a quantitative performance gain specifically to bidirectionality. The complete LoRA-ProtT5–Bi-Mamba configuration is therefore evaluated as a whole through group-aware OOF evaluation and controlled comparisons.

3.3.3 Multi-Seed Ensembling, Calibration, and Controlled Negative Results

After the basic sequence architecture had been established, we further investigated model variance and probability reliability. The final 1D expert averages the probabilities of three independently trained full-data models with equal weights; no seed was selected on test performance. The 15-member fold-checkpoint ensemble did not improve the OOF validation criteria and was therefore not used.

Following seed averaging, training OOF predictions were used to fit the monotonic Platt calibrator. This operation primarily adjusts the probability scale without altering the ranking of samples. The observed results were consistent with this role: the 1D Brier score decreased from 0.0445 to 0.0429 on Test1 and from 0.0491 to 0.0451 on Test2. After subsequent fusion with the 3D expert, the Brier scores were further reduced to 0.0410 and 0.0436, respectively. Calibration therefore plays a substantive role in the pipeline: it aligns the probability scales of the two experts before fusion.

Three intuitively attractive extensions were additionally evaluated but not adopted. Conservative positive-class sequence perturbation shifted predictions toward the positive class and degraded specificity and the Brier score. Length-specific calibration or thresholding was unstable

because of increased sampling variance within smaller strata and did not consistently outperform the global operating point. Finally, the 15-member fold ensemble did not outperform the three full-data models.

These variants did not improve the final 1D configuration. The three-seed full-data average with OOF calibration represents a simpler and more stable solution and reduces the risk of selective reporting or continued optimization against historically observed test sets.

Table 5: Selected controlled comparisons for the task-adaptive 1D sequence expert. OOF architecture rows and benchmark calibration rows are reported separately and should not be compared across evaluation splits.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC	Brier
Training OOF	Pooled ProtT5–TextCNN	0.9317	0.8586	0.9724	0.9436	NR
Training OOF	LoRA-ProtT5–Bi-Mamba	0.9300	0.8562	**0.9761**	**0.9533**	NR
Test1	Three-seed 1D, uncalibrated	–	–	0.9799	0.9644	0.0445
Test1	Three-seed 1D, OOF Platt	–	–	0.9799	0.9644	**0.0429**
Test2	Three-seed 1D, uncalibrated	–	–	0.9409	0.7831	0.0491
Test2	Three-seed 1D, OOF Platt	–	–	0.9409	0.7831	**0.0451**

3.4 Analysis of the pLDDT-Aware Multi-Scale 3D Structural Expert

Compared with the 1D expert, the principal methodological innovations of GHXTox lie in its structural expert. Previous studies have already demonstrated the value of predicted structures for toxicity prediction. PeptiTox combines ESM2 semantics with geometric descriptors extracted from predicted structures, whereas StrucToxNet further employs an equivariant graph neural network to preserve consistency under rigid transformations of the three-dimensional coordinates [6, 7]. For short peptides, however, the more fundamental question is which structural evidence should be trusted and at which spatial scale it should be used.

The GHXTox structural expert is organized around this question. ESMFold predictions are treated as uncertain structures whose pLDDT scores act as a continuous reliability variable throughout geometric message passing, residue-level gating, peptide-level gating, and training-time coordinate perturbation. The model then extends beyond Cα-level residue geometry by extracting local chemical pseudo-sites from full-atom PDB structures and, when conformer ensembles are available, incorporating statistics such as RMSF, distance variation, and contact occupancy through a conservative residual pathway.

The 3D contribution therefore lies in a hierarchical structural representation that organizes evidence by reliability and spatial scale, from residue geometry and local side-chain chemistry to sampled conformational variation.

3.4.1 Effect of Structural-Confidence Modeling

Structure-quality analysis first confirmed that pLDDT contains useful information: peptides with higher mean pLDDT generally exhibit stronger predictive performance. However, this does not imply that discarding low-confidence samples is an optimal modeling strategy. In a controlled high-pLDDT-only experiment, restricting training to high-confidence structures substantially reduced performance over the complete test domain, with MCC values of only 0.7232 on Test1 and 0.5593 on Test2.

Hard filtering removed difficult but informative samples along with uncertain structures.

GHXTox therefore uses a continuous confidence-aware strategy. Low-confidence edges remain in the graph but contribute weaker messages; residue-level gates modulate structural evi-

dence at individual positions; peptide-level gating controls the overall use of structural context; and stronger coordinate perturbations are applied to low-pLDDT residues during training.

This differs from approaches that analyze structure quality only after model inference. In GHXTox, pLDDT is an active component of representation learning, not a diagnostic variable appended after inference.

3.4.2 Contribution of Full-Atom Chemical Sites

A Cα-level residue graph effectively describes peptide compactness, nonlocal contacts, and residue neighborhoods but compresses chemically complex side chains into single residue nodes. To determine whether local spatial chemistry provides additional discriminative information, we introduced a full-atom-PDB-derived chemical-site residual on top of the frozen Cα-EGNN. A strict paired five-fold design was adopted: in each fold, the new model was initialized from the same control checkpoint, while the sequence branch, Cα-EGNN, fusion module, and classifier were frozen, leaving only the newly introduced chemical-site branch trainable.

Across the five CD-HIT homology-grouped folds, the complete chemical-site model increased mean BACC from 0.9153 to 0.9171 ($\Delta = +0.0018$), F1 from 0.8790 to 0.8814 ($\Delta = +0.0023$), MCC from 0.8312 to 0.8342 ($\Delta = +0.0030$), AUROC from 0.9687 to approximately 0.9692, and AUPRC from 0.9388 to 0.9401 ($\Delta = +0.0013$). More importantly, MCC, F1, and AUPRC improved in all five folds.

Although modest in magnitude, the improvements were consistent across folds. Because only the chemical-site branch was trainable, the result indicates that local side-chain chemistry provides a small but reproducible signal beyond residue-centered geometry.

Component-level ablations further support this interpretation. On the fixed fold-0 subset, removing the second chemical site, orientation invariants, either distance scale, explicit compatibility channels, pLDDT weighting, or zero initialization decreased MCC. Keeping only the raw-distance or only the normalized-distance representation reduced MCC by approximately 0.0059 and 0.0058, respectively, whereas removing interaction-type channels or pLDDT weighting each reduced MCC by approximately 0.0057.

PeptiTox demonstrated the utility of geometric information from predicted structures, whereas ToxiPep highlighted complementarity between local atomic chemistry and sequence context [4, 6]. GHXTox integrates these ideas within a confidence-aware 3D structural expert. The resulting interaction channels should nevertheless be interpreted as candidate spatial-chemical relationships available to the model, rather than confirmed hydrogen bonds, salt bridges, cation–π interactions, or disulfide bonds.

3.4.3 Sequence–Structure Interaction and Confidence Gating

Local geometric evidence becomes meaningful only in the appropriate sequence context. The same spatial distance or hydrophobic contact may have different implications depending on residue identity and sequence position. GHXTox therefore avoids simply concatenating sequence and structural vectors after peptide-level pooling. Instead, contextualized sequence states act as queries that selectively retrieve evidence from pLDDT-gated spatial residue representations.

This idea is conceptually related to the context–chemistry alignment emphasized by ToxiPep, but in GHXTox the interaction occurs within a predicted three-dimensional structural expert and is further regulated by both residue-level and peptide-level structural-confidence gates.

The residue-level gate modulates how strongly the structural representation of each residue contributes to the subsequent interaction, whereas the peptide-level gate controls the overall contribution of structural context. Structural information is therefore not switched off when confidence decreases; rather, its contribution is continuously attenuated at multiple levels.

Together with the negative result of hard high-pLDDT-only training described in Section 3.4.1, these observations suggest that the value of this design is the learnable compromise it establishes between preserving structural information and attenuating uncertain evidence.

Because no complete single-variable ablation of cross-attention and each confidence gate under otherwise identical conditions is available, the overall structural gain is not quantitatively decomposed among these components. The evidence therefore supports the reliability-aware interaction mechanism, comprising continuous pLDDT weighting, sequence–structure cross-attention, and hierarchical gating.

3.4.4 Effect of Conformer Statistics

A single ESMFold prediction provides only one deterministic structural anchor, whereas short peptides may exhibit substantial conformational heterogeneity. GHXTox does not treat every generated conformer as an additional labeled training example, nor does it repeatedly execute the complete EGNN for every conformation. Instead, conformer ensembles are compressed into statistics including RMSF, pairwise-distance variation, contact occupancy at 6, 8, and 10 Å, conformer RMSD, and radius of gyration, and these statistics are allowed to produce only a zero-initialized logit residual on top of the single-structure predictor.

The multi-source branch covers 2,795 of the 6,387 training peptides. At the unchanged 0.72/0.28 fusion and decision threshold, activating the multi-source residual increased training-OOF MCC from 0.87255 to 0.87445 and BACC from 0.93220 to 0.93292, while reducing the Brier score from 0.04316 to 0.04309. The MCC difference was +0.00190 under 5,000 class-stratified paired bootstrap resamples, with a 95% interval of [−0.00043, 0.00454] and a paired-randomization p value of 0.203. AUROC and AUPRC decreased by 0.00014 and 0.00015, respectively.

The gain remained limited: conformers showed substantial geometric variation, whereas expert probabilities changed only marginally. This suggests that the current bottleneck lies more in mapping conformational variation to toxicity labels than in generating geometrically distinct conformers. Conformer statistics produced small, inconsistent changes and were not used during benchmark inference. For peptides without valid conformer statistics, $\Delta z_{\text{conf}} = 0$, and the prediction falls back exactly to the single-structure pathway (the benchmark conformer ablation, alternative aggregators, and training-only optimization variants are detailed in the Supporting Information).

3.4.5 Overall Performance of the Structural Expert

After combining all components, the 3D-v2 structural expert achieved BACC, MCC, AUROC, and AUPRC values of 0.9208, 0.8464, 0.9711, and 0.9474 on Test1, compared with 0.9179, 0.8458, 0.9699, and 0.9426 for the historical Cα-only 3D-v1 model. Test1 therefore showed a consistent but modest improvement. Test2 exhibited a different trade-off: AUPRC increased from 0.6969 to 0.7241, whereas MCC decreased from 0.6320 to 0.6128.

This trade-off indicates that finer-grained structural modeling improves positive-class probability ranking but does not guarantee simultaneous gains in all metrics at a fixed operating point. It matters especially for Test2, whose 46 toxic peptides mean that a small number of threshold-adjacent samples can substantially alter MCC and F1.

The benefit of the 3D-v2 expert was dataset-dependent. Test1 improved across the reported metrics, whereas on Test2 AUPRC increased but the fixed-threshold MCC decreased; these gains were measured under group-aware training OOF evaluation.

Figure 5: Module-level ablation results for the 1D and 3D experts (A–F).

Table 6: Ablation analysis of the 3D structural expert. The first four rows use historical benchmark predictions; the final two rows are a paired five-fold OOF chemical-site experiment.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC
Test1	Historical Cα 3D-v1	0.9179	0.8458	0.9699	0.9426
Test1	pLDDT-aware multi-scale 3D-v2	**0.9208**	**0.8464**	**0.9711**	**0.9474**
Test2	Historical Cα 3D-v1	**0.8481**	**0.6320**	**0.9316**	0.6969
Test2	pLDDT-aware multi-scale 3D-v2	0.8453	0.6128	0.9295	**0.7241**
Training OOF	Frozen backbone, no chemical-site residual	0.9153	0.8312	0.9687	0.9388
Training OOF	Frozen backbone + chemical-site residual	**0.9171**	**0.8342**	**0.9692**	**0.9401**

3.5 Robustness, Statistical Uncertainty, and Applicability Analysis

Point estimates alone are insufficient for distinguishing a stable model improvement from variation caused by finite samples. We therefore use multi-seed averaging for the 1D expert and use paired, sample-level bootstrap resampling to compare the multi-source residual with its same-checkpoint zero-conformer OOF baseline. All comparisons preserve the pairing between the two predictions for each training peptide.

3.5.1 Multi-Seed Reproducibility

Three full-training 1D models initialized with seeds 42, 123, and 2025 are averaged before calibration. This ensemble is the final predictor; no seed is selected according to Test1 or Test2 performance. A 15-member ensemble of fold-specific checkpoints was also examined but not adopted because its members had each seen only approximately 80% of the training set and the aggregate did not improve the frozen validation criteria.

3.5.2 Paired Uncertainty and Point-Estimate Comparison

Paired-bootstrap and point-estimate comparisons involving the multi-source conformer variant and the preceding pilot model are reported in the Supporting Information; the variant was not part of the final benchmark predictor.

3.5.3 Selective Prediction and Risk–Coverage Analysis

The training-OOF analysis showed that proximity to the decision boundary provides an informative ranking of prediction risk. As coverage was reduced from 100% to 95%, 90%, 85%, and 80%, the accepted-sample error decreased from 5.12% to 3.66%, 2.92%, 2.41%, and 2.05%, respectively; MCC increased from 0.8733 to 0.9082, 0.9257, 0.9387, and 0.9481. The discrete OOF area under the risk–coverage curve was 0.01537. At the prespecified 90% operating point, the realized OOF coverage was 89.995% (5,748 of 6,387 peptides), which fixed the uncertainty cutoff at $d^{*}_{\mathrm{OOF}} = 0.47361765874159856$. Equal-count random rejection at this point retained an accepted error of 5.12%; the random-minus-selective difference was 2.19 percentage points (95% paired-bootstrap CI, 1.81–2.57). Random rejection produced no comparable improvement, indicating that the uncertainty ranking preferentially removed difficult samples.

The same cutoff was then transferred unchanged to the established benchmark sets. Test1 realized a coverage of 88.54% (997 of 1,126 peptides). Within the accepted subset, error decreased from the full-coverage value of 5.24% to 2.71%, MCC increased from 0.8705 to 0.9329, and BACC increased from 0.9323 to 0.9629. Sensitivity and specificity increased to 0.9397 and 0.9860, respectively. The full-minus-accepted error reduction was 2.53 percentage points (95% paired-bootstrap CI, 1.62–3.51), whereas equal-count random rejection retained an average error of 5.24% and an MCC of 0.8705.

Test2 showed the same direction at a realized coverage of 86.94% (506 of 582 peptides). Accepted-sample error decreased from 5.33% to 3.16%, MCC increased from 0.6673 to 0.7781, and BACC increased from 0.8618 to 0.8983. Sensitivity and specificity reached 0.8158 and 0.9808. The corresponding error reduction was 2.16 percentage points (95% paired-bootstrap CI, 0.99–3.52), while the equal-count random control retained an average error of 5.33% and an MCC of 0.6670. Exact operating-point values are reported in Table 7.

Table 7: The OOF-frozen selective policy identifies lower-risk accepted subsets on both established benchmarks. Random-rejection values are means over 5,000 masks with the same accepted count as the frozen policy. The binary probabilities, fusion rule, and decision threshold are unchanged.

Dataset	Method	Coverage (%)	Error (%)	MCC	BACC	SN	SP
Test1	Full binary prediction	100.00	5.24	0.8705	0.9323	0.8969	0.9677
	Frozen selective prediction	88.54	2.71	0.9329	0.9629	0.9397	0.9860
	Equal-count random rejection	88.54	5.24	0.8705	0.9323	0.8969	0.9677
Test2	Full binary prediction	100.00	5.33	0.6673	0.8618	0.7609	0.9627
	Frozen selective prediction	86.94	3.16	0.7781	0.8983	0.8158	0.9808
	Equal-count random rejection	86.94	5.33	0.6670	0.8617	0.7607	0.9627

The transfer rule did not use a benchmark-specific quantile, desired coverage, class prevalence, or ground-truth label. The naturally different Test1 and Test2 coverages are therefore consequences of applying one OOF-derived cutoff, not of retuning the abstention rate. These results support a risk-aware toxicity-screening workflow in which boundary-proximal predictions are flagged for additional review. They do not constitute validation on a newly collected independent dataset, and the original binary prediction remains available for every abstained peptide.

Figure 6 reserves the combined OOF and benchmark presentation; only its caption and placement are fixed at this stage.

Figure 6: OOF-frozen decision-boundary proximity identifies lower-risk accepted subsets and transfers without benchmark-specific tuning. (A) Accepted-sample error and (B) MCC across 100%, 95%, 90%, 85%, and 80% coverage on group-aware training OOF predictions. The frozen uncertainty ranking is compared with equal-count random rejection, and the vertical dashed line marks the prespecified 90% OOF operating point used to obtain the fixed cutoff. Shaded regions denote 95% intervals from 5,000 label-stratified paired-bootstrap replicates. (C, D) Full-coverage, equal-count random-rejection, and frozen-policy performance on Test1 and Test2 at the naturally realized benchmark coverages. No benchmark-specific quantile was recomputed, and benchmark labels were accessed only after the accept/abstain decisions had been generated, solely for evaluation.

3.6 Interpretability and Representative Case Studies

The interpretation experiments used the model and the prespecified case-selection rule described above. The analysis included 460 residue occlusions across 16 peptides, unperturbed confidence gates for all 1,708 benchmark records, 445 individual chemical-site deletions, 121 category-level deletions, and 424 residue-level site deletions. Before perturbation, baseline probabilities were recomputed and compared with the stored prediction vectors. The maximum absolute discrepancy for the 1D expert within the case cohort was 3.45×10^{-4}; the corresponding 3D discrepancies were 5.49×10^{-4} on Test1 and 1.14×10^{-5} on Test2. None of these differences changed a standalone 3D or fused decision. The perturbation baselines therefore preserved the

stored decision states, with the larger Test1 discrepancy reported as bounded cross-environment numerical drift.

3.6.1 Residue-Level Interpretation of the 1D Expert

The residue-occlusion profiles were heterogeneous across peptides and seeds. Across the 16 prespecified cases, the mean pairwise Spearman correlation of absolute per-residue logit effects was 0.118 (case-level range, −0.077 to 0.422), whereas the mean residue-wise sign-agreement fraction was 0.740 (range, 0.667–0.796). Direction was therefore more stable across seeds than the fine-grained ranking of residue magnitudes. The reserved maps emphasize local effects with consistent signs rather than implying a single stable ranking across all residues.

Several cases nevertheless contained large, directionally consistent effects. For the Test2 toxic peptide `sequence_34_34`, occlusion of Asn9 reduced the calibrated 1D probability from 0.5447 to 0.0412 ($\Delta p_{1D} = +0.5035$), with all three seeds agreeing on the sign. In the Test1 toxic case `peptide_199`, occlusion of Ser24 reduced the score from 0.3687 to 0.1114 ($\Delta p_{1D} = +0.2573$), again with complete sign agreement. Effects were not uniformly toxicity-supporting: in the Test2 toxic case `sequence_23_23`, occlusion of Ile5 increased the 1D probability from 0.2807 to 0.4819 ($\Delta p_{1D} = -0.2013$).

These cases illustrate large, context-dependent sensitivities in the 1D output; they do not identify experimentally validated toxic residues.

The dataset-specific layouts are reserved in Figures 7 and 8, respectively.

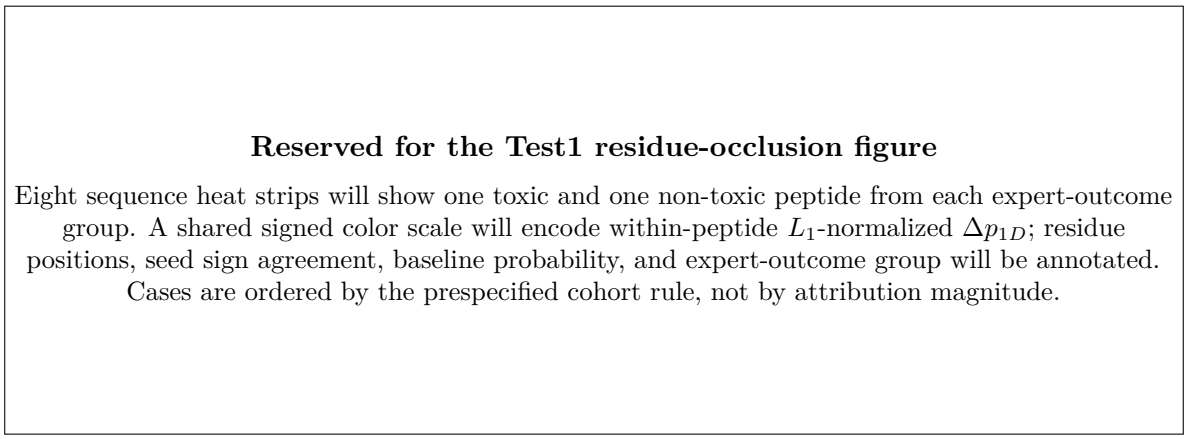


Figure 7: Reserved layout for Test1 residue-level analysis of the 1D expert. Each row will represent one independently selected peptide and each cell one residue. Positive values will indicate that zeroing the post-contextual residue vector lowers the calibrated 1D toxicity probability, whereas negative values will indicate an increase. The common color normalization will permit comparison of direction and relative within-peptide magnitude; it will not imply that equal colors correspond to equal biological effects across peptides. Seed agreement will distinguish reproducible directions from effects dominated by one checkpoint.

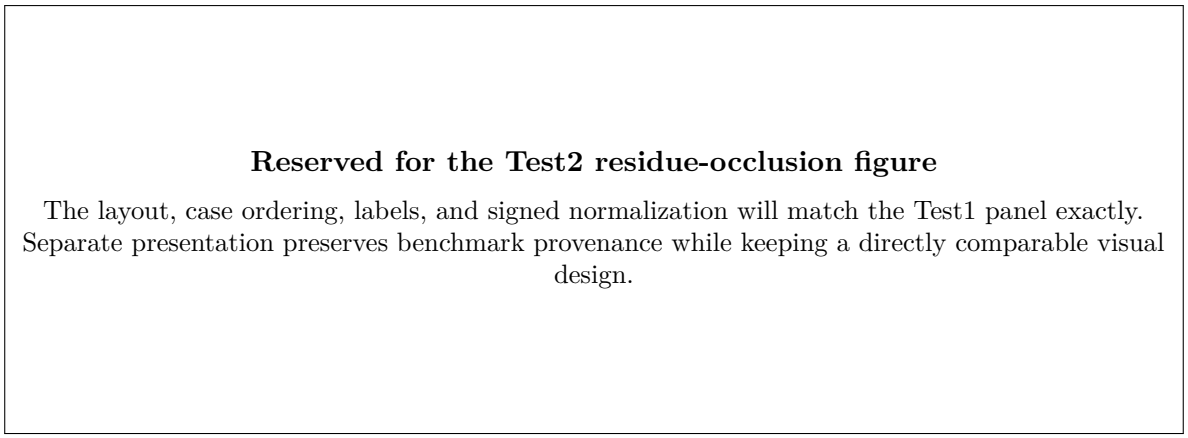


Figure 8: Reserved layout for Test2 residue-level analysis of the 1D expert. The eight rows will cover both labels in all four expert-outcome groups. Residue-wise probability shifts will be computed with the three seeds, the original inference padding context, and the fixed OOF Platt transform. Test2 comprises exact sequence matches to a subset of Test1, so matched records may yield similar maps. Such agreement will be treated as a reproducibility check, not as independent external validation.

3.6.2 Structural Confidence and Chemical-Site Sensitivity

Mean pLDDT was strongly associated with the learned global structural gate. Spearman correlations were 0.960 on Test1 (95% peptide-bootstrap interval, 0.953–0.966) and 0.950 on Test2 (0.937–0.959); the pooled descriptive value was 0.957 (0.951–0.961). In contrast, the mean within-peptide correlation between residue pLDDT and node gates was small: 0.049 on Test1 (0.030–0.068) and 0.064 on Test2 (0.036–0.092). Thus, global gating closely tracked confidence at the peptide level, whereas the weak local correlations are consistent with residue gates also reflecting geometric and contextual features.

Most individual-site deletions produced small changes, but the distribution had a pronounced upper tail. Across 445 deletions, the median and mean absolute changes in p_{3D} were 0.00048 and 0.00997, respectively; the 95th percentile was 0.0570 and the maximum was 0.2994. Nine individual deletions moved the recomputed fused probability across the decision threshold when the 1D probability was held fixed. At category level, hydrophobic-site deletion had the largest mean absolute change among the eight modeled categories (0.126 across 16 case-level deletions), followed by aromatic sites (0.0687), whereas sulfur-site deletion had the smallest mean absolute change (0.00179). Because the analysis used the selected 16-peptide cohort, these category-level estimates are descriptive, not population effects.

The largest individual-site effect occurred at the aromatic/hydrophobic Phe8 site in the matched cache-covered cases `peptide_846` and `sequence_388_388`: deletion increased p_{3D} from approximately 0.2405 to 0.5399, giving $\Delta p_{3D} \approx -0.2994$. Conversely, deletion of the hydrophobic/conformational Pro19 site in `peptide_199` decreased p_{3D} from 0.3443 to 0.1163 ($\Delta p_{3D} = +0.2280$).

The opposing directions show that the deletion score depends on the surrounding peptide context and cannot be read as an intrinsic toxicity label for a functional group.

Figure 9 reserves a layout for these scale-specific associations and perturbation effects.

Reserved for the structural-sensitivity figure

(A) Mean pLDDT versus the learned global gate for Test1 and Test2, with dataset-specific Spearman coefficients and bootstrap intervals. (B) Distribution of within-peptide pLDDT–node-gate correlations. (C) Signed and absolute Δp_{3D} for category-level deletions. (D) Largest individual-site deletions, labeled by residue, position, chemical-site type, conformer availability, and induced fused shift.

Figure 9: Reserved layout for structural-confidence and chemical-site sensitivity analysis. (A) Association between mean pLDDT and the global gate. (B) Distribution of within-peptide pLDDT–node-gate correlations. (C) Comparison of the eight modeled chemical categories. (D) Largest probability-increasing and probability-decreasing site-deletion effects. The deletion panels use the 3D expert and calculate Δp_{fused} with p_{1D} fixed; the reported values denote model sensitivity, not experimentally verified toxicophores.

3.6.3 Representative Expert-Disagreement Cases

The aggregate disagreement pattern is reported in Section 3.2.3. Here, we focus on representative cases in which the two experts provide opposing evidence and the fixed fusion either rescues or reverses a correct standalone decision.

The toxic Test2 peptide `sequence_23_23` illustrates a structural rescue. Its 1D probability was 0.2806, below the standalone sequence threshold, whereas the 3D probability was 0.9954. The fixed 0.72/0.28 fusion yielded $p_{\text{fused}} = 0.4808$, moving the final prediction just above the predeclared threshold and restoring the correct toxic classification. The residue-occlusion analysis also showed that zeroing the contextual vector at Ile5 increased the 1D score, underscoring that the sequence evidence in this case was both weak and context dependent.

The toxic peptide `sequence_34_34` shows the opposite outcome. Its 1D probability of 0.5447 was correct at the standalone threshold, but its 3D probability was 3.0×10^{-5}. Fusion reduced the probability to 0.3922 and produced a false negative. In the 1D sensitivity analysis, occluding Asn9 reduced the calibrated sequence probability by 0.5035 with complete cross-seed sign agreement, indicating that strong sequence evidence can be overridden when the structural expert supplies sufficiently strong opposing evidence.

These cases do not establish a biological mechanism. They demonstrate the operational consequence of fixed late fusion: complementary evidence can correct an expert-specific error, but the same rule can also regress an otherwise correct decision. Figure 10 reserves the case-level presentation without repeating the aggregate counts shown in Figure 4.

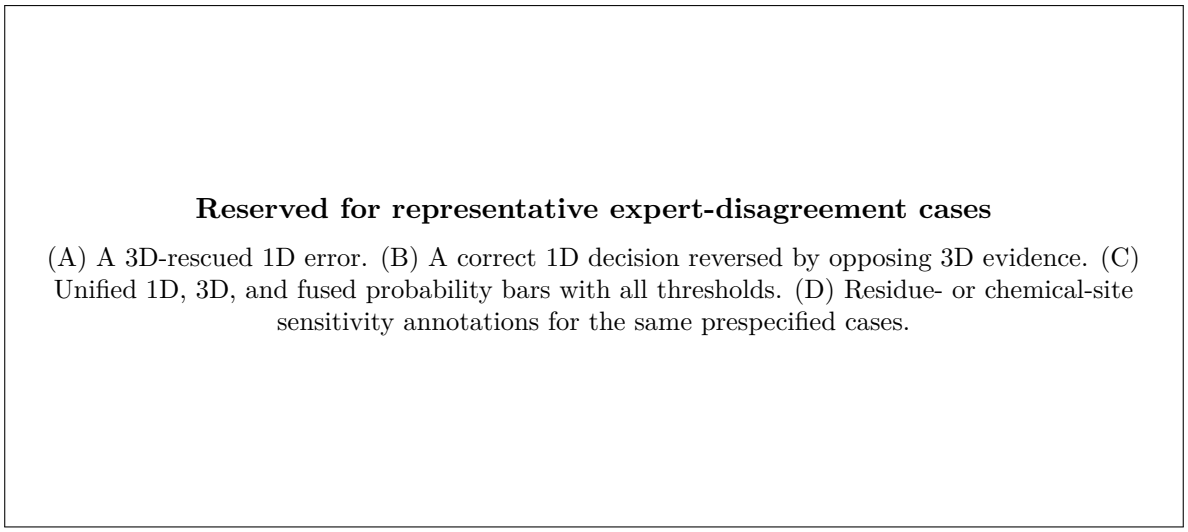


Figure 10: Representative disagreements show that fixed late fusion can either rescue an expert-specific error or override an otherwise correct prediction. (A) A peptide for which the 3D expert restores a toxic decision missed by the 1D expert. (B) A peptide for which opposing structural evidence reverses a correct 1D prediction. (C) The three probability streams shown against their decision thresholds. (D) Model-sensitivity annotations for the same cases. Residue occlusion and chemical-site deletion quantify sensitivity of the predictor and are not interpreted as causal biological effects.

3.7 Limitations and Future Directions

Several limitations remain. Test1 and Test2 were observed during earlier model development, so they are frozen historical benchmarks rather than blinded external test sets, and Test2 contains only 46 toxic peptides, so its SN, BACC, and MCC are sensitive to a few prediction changes. Stronger generalization evidence will require a new dataset evaluated under a prefrozen protocol.

Structural predictions remain uncertain: pLDDT reflects predicted confidence rather than experimental accuracy or conformational flexibility, and full-atom-derived chemical sites are candidate spatial-chemical relationships rather than confirmed bonds. Conformer coverage is 43.16% of Test1 and 48.80% of Test2, with generator identity partly coupled to peptide length; the conformer pathway is therefore a conditional refinement, not a universal source of improvement. More flexible residual variants did not yield a stable multimetric improvement, and the single sample-level toxicity label does not specify which conformer or microstate should carry the activity.

The two experts are trained separately and combined with a fixed fusion weight that cannot adapt per sample. Interpretability analyses are limited in scope: residue occlusion covers a 16-peptide cohort with modest cross-seed rank agreement, and chemical-site deletion operates on predicted-structure features. Future work should therefore prioritize a new independent external dataset, better-controlled same-sample conformer sets, and representation learning for the conformer pathway rather than further fixed aggregators; orthogonal attribution methods and, where feasible, mutational or toxicity experiments could then test the resulting hypotheses.

4 CONCLUSIONS

GHXTox combines a task-adaptive LoRA-ProtT5–Bi-Mamba sequence expert with a pLDDT-aware multi-scale structural expert through a fixed probability fusion protocol. The structural expert integrates residue geometry, full-atom-derived chemical sites, and sequence-

conditioned structural evidence, while multi-conformer information is restricted to training and controlled ablation. The final predictor uses the structural expert's single-structure base probability. The 1D/3D weights (0.72/0.28) and the decision threshold (0.47022256531284556) were fixed from training OOF predictions before benchmark evaluation.

On the frozen historical Test1 and Test2 benchmarks, the model reaches MCC values of 0.8705 and 0.6673 and AUPRC values of 0.9674 and 0.8046, and the final fusion improves AUROC, AUPRC, and the Brier score relative to the calibrated 1D stream. The 3D expert correctly classified 36.92% and 37.50% of the respective Test1 and Test2 errors made by the 1D expert, with most of these structural rescues occurring among non-toxic peptides. An OOF-frozen selective policy further reduced accepted-sample error to 2.71% and 3.16% at realized coverages of 88.54% and 86.94%, respectively, without benchmark-specific cutoff selection. Conformer-derived information produced only modest and non-uniform improvements, indicating that mapping conformational variation to toxicity-relevant evidence remains an important limitation. The interpretability analyses describe model sensitivity rather than causal mechanisms.

Several limitations remain: both benchmarks were observed during earlier model development, the selective rule has not been prospectively validated, and the structures are predicted rather than experimental. Future work should prioritize a new independent external dataset, better-controlled same-sample conformer sets, and representation learning for the conformer pathway, followed where feasible by experimental validation.

Data and Software Availability

The source code, configuration files, fold manifests, and inference scripts used in this study are available at `https://github.com/Suiyumanbu/GHXTox`. A versioned archival release containing the exact commit identifier, environment manifest, frozen model parameters, calibration coefficients, fusion weights, full-precision decision threshold, and sample-level prediction files will accompany the final publication. Raw peptide records were obtained from the public databases described in Section 2.1. Where redistribution is restricted by the original database terms, accession lists and preprocessing scripts are provided instead of redistributing the source records.

Supporting Information

A Supporting Information package containing detailed dataset documentation, fold-wise results, hyperparameter settings, detailed bootstrap and randomization procedures, seed-level reproducibility results, training-OOF conformer information-ceiling diagnostics, additional ablation experiments, controlled benchmark conformer ablations and paired point-estimate comparisons, subgroup analyses, statistical outputs, extended interpretability case studies, and conformer-covered sensitivity cases will accompany the submitted manuscript.

CRediT Authorship Contribution Statement

Acknowledgements

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Markus Muttenthaler, Glenn F. King, David J. Adams, and Paul F. Alewood. Trends in peptide drug discovery. *Nature Reviews Drug Discovery*, 20(4):309–325, 2021.

[2] Shengli Zhang, Jingyi Ren, and Yunyun Liang. An innovative peptide toxicity prediction model based on multi-scale convolutional neural network and residual connection. *Bioinformatics*, 41(10):btaf462, 2025.

[3] Yunyun Liang and Chenxia Wang. ToxPLTC: Peptide toxicity prediction by integrating a pre-trained T5 protein language model and text convolutional neural network. *Journal of Chemical Information and Modeling*, 66(7):4058–4074, 2026.

[4] Jiahui Guan, Peilin Xie, Dian Meng, Lantian Yao, Dan Yu, Ying-Chih Chiang, Tzong-Yi Lee, and Junwen Wang. ToxiPep: Peptide toxicity prediction via fusion of context-aware representation and atomic-level graph. *Computational and Structural Biotechnology Journal*, 27:2347–2358, 2025.

[5] Duong Thanh Tran, Nhat Truong Pham, Nguyen Doan Hieu Nguyen, Leyi Wei, and Balachandran Manavalan. HyPepTox-Fuse: An interpretable hybrid framework for accurate peptide toxicity prediction fusing protein language model-based embeddings with conventional descriptors. *Journal of Pharmaceutical Analysis*, 15(8):101410, 2025.

[6] Yanling Wang, Na Li, Xiao Wang, Feng Cao, Shuwen Xiong, and Leyi Wei. Integrating protein language models and geometric deep learning for peptide toxicity prediction. *Journal of Chemical Information and Modeling*, 65(14):7800–7810, 2025.

[7] Shihu Jiao, Xiucai Ye, Tetsuya Sakurai, Quan Zou, Wu Han, and Chao Zhan. Integration of pre-trained protein language models with equivariant graph neural networks for peptide toxicity prediction. *BMC Biology*, 23:229, 2025.

[8] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, et al. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022.

[9] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[10] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling*, 2024.

[11] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.

[12] John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.

[13] Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, et al. ColabFold: Making protein folding accessible to all. *Nature Methods*, 19(6):679–682, 2022.

[14] Osama Abdin and Philip M. Kim. Direct conformational sampling from peptide energy landscapes through hypernetwork-conditioned diffusion. *Nature Machine Intelligence*, 6(7):775–786, 2024.

[15] Shengli Zhang and Xinyi Zhang. NeuroPpred-PSCG: A multimodal framework using ProtT5 and structural features for neuropeptide prediction based on gated cross-attention mechanism and multiscale gated convolution. *Journal of Chemical Information and Modeling*, 2026.

[16] Shihu Jiao, Xiucai Ye, Tetsuya Sakurai, Quan Zou, and Ran Liu. Integrated convolution and self-attention for improving peptide toxicity prediction. *Bioinformatics*, 40(5):btae297, 2024.

[17] Vitali Morozov, Carlos H. M. Rodrigues, and David B. Ascher. CSM-Toxin: A web-server for predicting protein toxicity. *Pharmaceutics*, 15(2):431, 2023.

[18] Nishu Sharma, Leimarembi Devi Naorem, Shilpi Jain, and Gajendra P. S. Raghava. ToxinPred2: An improved method for predicting toxicity of proteins. *Briefings in Bioinformatics*, 23(5):bbac174, 2022.

[19] Lesong Wei, Xiucai Ye, Yuyang Xue, Tetsuya Sakurai, and Leyi Wei. ATSE: A peptide toxicity predictor by exploiting structural and evolutionary information based on graph neural network and attention mechanism. *Briefings in Bioinformatics*, 22(5):bbab041, 2021.

[20] The UniProt Consortium. UniProt: A worldwide hub of protein knowledge. *Nucleic Acids Research*, 47(D1):D506–D515, 2019.

[21] Limin Fu, Beifang Niu, Zhengwei Zhu, Sitao Wu, and Weizhong Li. CD-HIT: Accelerated for clustering the next-generation sequencing data. *Bioinformatics*, 28(23):3150–3152, 2012.

[22] Akanksha S. Rathore, S. Choudhury, A. Arora, P. Tijare, and Gajendra P. S. Raghava. ToxinPred 3.0: An improved method for predicting the toxicity of peptides. *Computers in Biology and Medicine*, 179:108926, 2024.

[23] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter L. Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, Cambridge, MA, 1999.

[24] Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. $E(n)$ equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9323–9332, 2021.

[25] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[26] Brian W. Matthews. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. *Biochimica et Biophysica Acta (BBA) – Protein Structure*, 405(2):442–451, 1975.

[27] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[28] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993.

[29] Markus Ojala and Gemma C. Garriga. Permutation tests for studying classifier performance. *Journal of Machine Learning Research*, 11:1833–1863, 2010.